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Lovold

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(54) **ALL-TERRAIN VEHICLE**

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(56) **References Cited**

U.S. PATENT DOCUMENTS

610,132 A 8/1898 Goodridge
2,309,432 A 1/1943 Alden
(Continued)

FOREIGN PATENT DOCUMENTS

AU 2013317853 A1 4/2015
AU 2015266694 A1 1/2017
(Continued)

OTHER PUBLICATIONS

Annex to Form PCT/ISA/206 Communication Relating to the
Results of the Partial International Search issued by the European
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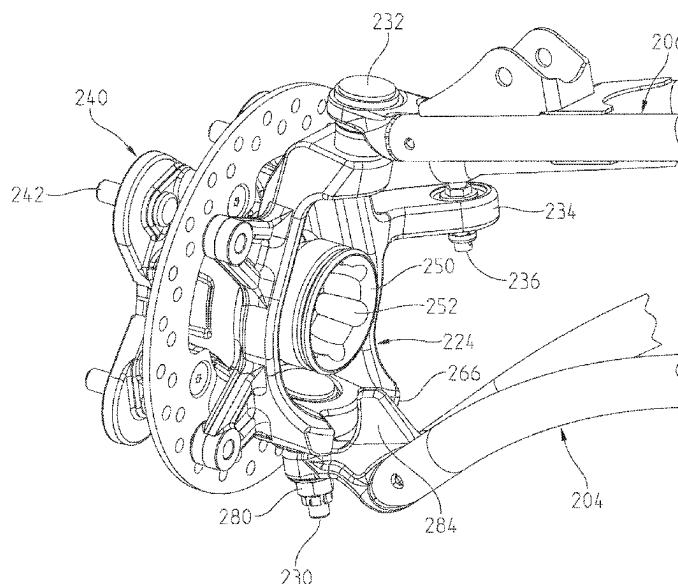
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(57) **ABSTRACT**

An ATV is shown having a steering system comprised of a
power steering unit having drive and driven pitman arms
coupled to a drag link. The drive pitman arm is laterally
offset from a vehicle centerline and is driven by the power
steering unit. An alternate power steering system includes a
pack and pinion subassembly coupled to a power steering
motor, which then couples to steering arms of the ATV.

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(56)**References Cited****U.S. PATENT DOCUMENTS**

2,553,795 A	5/1951	Staude	4,987,965 A	1/1991	Bourret	
2,672,103 A	3/1954	Hohmes	5,010,970 A	4/1991	Yamamoto	
2,834,608 A	5/1958	Wixson	5,020,616 A	6/1991	Yagi et al.	
3,259,211 A	7/1966	Ryskamp	5,021,721 A	6/1991	Oshita et al.	
3,336,074 A	8/1967	Barnes et al.	5,027,915 A	7/1991	Suzuki et al.	
RE26,775 E	1/1970	Smieja	5,036,939 A	8/1991	Johnson et al.	
3,563,564 A *	2/1971	Bartkowiak F16C 11/0604 403/126	5,042,835 A	8/1991	Burns	
3,688,856 A	9/1972	Boehm et al.	5,052,528 A *	10/1991	Sullivan	B60G 7/04 267/221
3,712,398 A	1/1973	Althaus	5,057,062 A	10/1991	Yamasaki et al.	
3,712,416 A	1/1973	Swanson et al.	5,060,745 A	10/1991	Yasui et al.	
3,814,201 A	6/1974	O'Brien	5,078,225 A	1/1992	Ohmura et al.	
3,833,253 A	9/1974	Butler	5,078,229 A	1/1992	Kikuchi et al.	
3,840,082 A	10/1974	Olson	5,107,951 A	4/1992	Kawamura	
3,856,323 A	12/1974	Arning et al.	5,205,371 A	4/1993	Karnopp	
3,858,902 A	1/1975	Howells et al.	5,251,713 A	10/1993	Enokimoto	
3,917,310 A	11/1975	Mitsubishi	5,253,730 A	10/1993	Hayashi et al.	
3,933,213 A	1/1976	Trowbridge	5,327,989 A	7/1994	Furuhashi et al.	
3,966,014 A	6/1976	Gowing	5,370,198 A	12/1994	Karpik	
3,969,895 A	7/1976	Krizman	5,393,194 A	2/1995	Smith	
RE28,922 E	8/1976	Lloyd, III	5,429,290 A	7/1995	Greene, Jr.	
3,977,493 A	8/1976	Richardson	5,588,660 A *	12/1996	Paddison	B62D 7/18 280/93.512
4,027,892 A	6/1977	Parks	5,692,579 A	12/1997	Peppel et al.	
4,045,075 A	8/1977	Pulver	5,692,759 A	12/1997	Flynn	
4,079,807 A	3/1978	Hornagold et al.	5,697,332 A	12/1997	Asai et al.	
4,089,542 A	5/1978	Westerman	5,704,866 A	1/1998	Pritchard et al.	
4,098,414 A	7/1978	Abiera	5,708,238 A	1/1998	Asao et al.	
4,222,453 A	9/1980	Fixsen et al.	5,727,643 A	3/1998	Kawano et al.	
4,254,746 A	3/1981	Chiba et al.	5,753,300 A	5/1998	Wessels et al.	
4,314,618 A	2/1982	Tamura	5,779,272 A	7/1998	Panek et al.	
4,339,156 A	7/1982	Livesay	5,842,732 A	12/1998	Daggett et al.	
4,340,124 A	7/1982	Leonard	5,860,486 A	1/1999	Boivin et al.	
4,407,386 A	10/1983	Yasui et al.	5,863,277 A	1/1999	Melbourne	
4,442,913 A	4/1984	Grinde	5,881,834 A	3/1999	Karpik	
4,561,323 A	12/1985	Stromberg	5,887,671 A	3/1999	Yuki et al.	
4,577,716 A	3/1986	Norton	5,904,216 A	5/1999	Furusawa	
4,626,024 A	12/1986	Swann	5,944,134 A	8/1999	Peppel et al.	
4,627,512 A	12/1986	Clohesy	5,947,217 A	9/1999	Snare et al.	
4,653,773 A *	3/1987	Graft B62D 5/061 280/93.5	5,950,750 A	9/1999	Dong et al.	
4,657,271 A	4/1987	Salmon	5,954,364 A	9/1999	Nechushtan	
4,679,450 A	7/1987	Hayakawa et al.	5,971,434 A	10/1999	Neufeld et al.	
4,686,433 A	8/1987	Shimizu	6,029,750 A	2/2000	Carrier	
4,695,089 A	9/1987	Fukutomi et al.	6,070,689 A	6/2000	Tanaka et al.	
4,700,982 A	10/1987	Kuraoka et al.	6,092,877 A	7/2000	Rasidescu et al.	
4,782,912 A	11/1988	Wandless	6,126,227 A	10/2000	Bitelli	
4,821,825 A	4/1989	Somerton-Rayner	6,149,540 A	11/2000	Johnson et al.	
4,858,722 A	8/1989	Abbe et al.	6,161,853 A	12/2000	Jung	
4,917,207 A	4/1990	Yasui et al.	6,161,908 A	12/2000	Takayama et al.	
4,934,737 A	6/1990	Nakatsuka	6,199,894 B1	3/2001	Anderson	
			6,206,124 B1	3/2001	Mallette et al.	
			6,227,323 B1	5/2001	Ashida	
			6,283,241 B1	9/2001	Kubota	
			6,293,605 B2	9/2001	Neubrand	
			6,321,864 B1	11/2001	Forbes	
			6,343,578 B1	2/2002	Kerkau et al.	
			6,378,645 B1	4/2002	Uegane et al.	
			6,379,411 B1	4/2002	Turner et al.	
			6,390,219 B1	5/2002	Vaisanen	
			6,398,240 B1 *	6/2002	Taylor	B60T 1/065 280/93.511
			6,412,797 B1	7/2002	Park	
			6,416,108 B1	7/2002	Elswick	
			6,431,569 B2	8/2002	Handa	
			6,461,208 B2	10/2002	Suzuki et al.	
			6,467,561 B1	10/2002	Boivin et al.	
			6,467,787 B1	10/2002	Marsh	
			6,478,098 B2	11/2002	Boivin et al.	
			6,520,284 B2	2/2003	Spannbauer et al.	
			6,551,385 B2	4/2003	Turner et al.	
			6,554,665 B1	4/2003	Kaneko	
			RE38,124 E	5/2003	Mallette et al.	
			6,557,530 B1	5/2003	Benson et al.	
			6,557,922 B1	5/2003	Hommel	
			6,568,030 B1	5/2003	Watanabe et al.	
			6,595,309 B1	7/2003	Savage et al.	
			6,622,968 B1	9/2003	St et al.	
			6,626,258 B1	9/2003	Forbes	
			6,631,778 B2	10/2003	Mallette	

(56)

References Cited

U.S. PATENT DOCUMENTS

6,695,083 B2	2/2004	Nakamura et al.	7,677,599 B2	3/2010	Kuerten et al.
6,695,566 B2	2/2004	Rodriguez Navio	7,677,634 B2	3/2010	Flynn
6,715,460 B2	4/2004	Ashida et al.	7,708,103 B2	5/2010	Okuyama et al.
6,715,575 B2	4/2004	Karpik	7,717,495 B2	5/2010	Leonard et al.
6,725,958 B2	4/2004	Ashida et al.	7,753,162 B2	7/2010	Shiozawa et al.
6,725,962 B1	4/2004	Fukuda	7,761,217 B2	7/2010	Waters et al.
6,745,862 B2	6/2004	Morii et al.	7,762,567 B2	7/2010	Marur
6,752,235 B1	6/2004	Bell et al.	7,775,318 B2	8/2010	Okada et al.
6,755,271 B1	6/2004	Berg	7,782,000 B2	8/2010	Suzuki
6,767,022 B1	7/2004	Chevalier	7,795,602 B2	9/2010	Leonard et al.
6,786,526 B1	9/2004	Blalock	7,819,220 B2	10/2010	Sunsdahl et al.
6,796,395 B1	9/2004	Berg et al.	7,854,290 B1	12/2010	McKoskey
6,799,779 B2	10/2004	Shibayama	7,871,106 B2	1/2011	Leonard et al.
6,799,781 B2	10/2004	Rasidescu et al.	RE42,086 E	2/2011	Saito et al.
6,823,834 B2	11/2004	Benson et al.	7,882,817 B2	2/2011	Yamazaki
6,851,691 B2	2/2005	Rasidescu et al.	7,896,422 B2	3/2011	Willard et al.
6,916,142 B2	7/2005	Hansen et al.	7,900,743 B2	3/2011	Kaku et al.
6,923,507 B1	8/2005	Billberg et al.	7,922,201 B2	4/2011	Kuerten et al.
6,926,108 B1	8/2005	Polakowski et al.	7,946,380 B2	5/2011	Yamamoto et al.
6,942,050 B1	9/2005	Honkala et al.	7,950,486 B2	5/2011	Van et al.
6,966,399 B2	11/2005	Tanigaki et al.	7,967,365 B2	6/2011	Gerhardt et al.
6,978,857 B2	12/2005	Korenjak	7,971,904 B2	7/2011	David
7,040,437 B1	5/2006	Fredrickson et al.	7,992,671 B2	8/2011	Okada et al.
7,047,924 B1	5/2006	Waters et al.	8,029,021 B2	10/2011	Leonard et al.
7,055,454 B1	6/2006	Whiting et al.	8,104,524 B2	1/2012	Manesh et al.
7,063,057 B1	6/2006	Waters et al.	8,122,993 B2	2/2012	Ripley et al.
7,070,012 B2	7/2006	Fecteau	8,176,957 B2	5/2012	Manesh et al.
7,077,233 B2	7/2006	Hasegawa	8,191,952 B2	6/2012	Mokhtari et al.
7,096,988 B2	8/2006	Moriyama	8,205,910 B2	6/2012	Leonard et al.
7,124,853 B1	10/2006	Kole, Jr.	8,240,748 B2	8/2012	Chapman
7,125,134 B1	10/2006	Hedlund et al.	8,328,235 B2	12/2012	Schneider et al.
7,165,702 B1	1/2007	Billberg	8,353,534 B2	1/2013	Arnold et al.
7,182,165 B1	2/2007	Keinath et al.	8,376,078 B2	2/2013	Hiddema
7,182,169 B2	2/2007	Suzuki	8,382,130 B2	2/2013	Nakamura
7,185,732 B2	3/2007	Saito et al.	8,408,354 B2	4/2013	Nozaki et al.
7,210,547 B2	5/2007	Nojima	8,454,041 B2	6/2013	Fujii et al.
7,216,733 B2	5/2007	Iwami et al.	8,517,135 B2	8/2013	Schapf et al.
7,222,882 B2	5/2007	Boucher	8,517,140 B2	8/2013	West et al.
7,237,789 B1	7/2007	Herman	8,544,582 B2	10/2013	Kaku et al.
7,237,802 B2	7/2007	Rose et al.	8,544,961 B2 *	10/2013	Bubulka B60B 35/003 301/124.1
7,237,803 B2	7/2007	Nguyen	8,596,709 B2	12/2013	Ugolini
7,239,032 B1	7/2007	Wilson et al.	8,602,159 B2	12/2013	Harris et al.
7,240,472 B2	7/2007	Evers	8,613,335 B2	12/2013	Deckard et al.
7,240,960 B2	7/2007	Fallis et al.	8,640,814 B2	2/2014	Deckard et al.
7,249,652 B2	7/2007	Wolf	8,651,800 B2	2/2014	Li
7,331,896 B1	2/2008	Kroppe	8,662,216 B2	3/2014	Roucka
7,370,724 B2	5/2008	Saito et al.	8,662,224 B2	3/2014	Shiokawa et al.
7,374,012 B2	5/2008	Inui et al.	8,746,719 B2	6/2014	Safranski et al.
7,380,622 B2	6/2008	Shimizu	8,827,028 B2	9/2014	Sunsdahl et al.
7,380,805 B1	6/2008	Turner	8,833,502 B2	9/2014	Sellers
7,395,890 B2	7/2008	Visscher	8,893,835 B2	11/2014	Nam et al.
7,410,182 B1	8/2008	Giese et al.	8,944,465 B2	2/2015	Shinbori
7,431,024 B2	10/2008	Buchwitz et al.	8,967,666 B2	3/2015	Iardella et al.
7,434,656 B2	10/2008	Yasuda et al.	8,978,819 B2	3/2015	Fujimoto et al.
7,444,236 B2	10/2008	Wiles	8,979,124 B2	3/2015	Shinbori et al.
7,455,141 B2	11/2008	Hildebrand	9,004,224 B2	4/2015	Uzawa
7,458,593 B2	12/2008	Saito et al.	9,096,289 B2	8/2015	Hedlund et al.
7,481,293 B2	1/2009	Ogawa et al.	9,150,182 B1	10/2015	Schlangen et al.
7,485,984 B2	2/2009	Fulks et al.	9,248,861 B2	2/2016	Koren et al.
7,497,299 B2	3/2009	Kobayashi	9,248,886 B1	2/2016	Giese
7,497,471 B2	3/2009	Kobayashi	9,358,836 B2	6/2016	David et al.
7,500,709 B2	3/2009	Heselhaus	9,381,803 B2	7/2016	Galsworthy et al.
7,523,799 B2	4/2009	Yoshihara	9,440,671 B2	9/2016	Schlangen et al.
7,540,511 B2	6/2009	Saito et al.	9,506,407 B2	11/2016	Blake
7,559,596 B2	7/2009	Garska et al.	9,540,072 B2	1/2017	Hedlund et al.
7,562,736 B2	7/2009	Eguchi et al.	9,592,782 B2	3/2017	Raska et al.
7,565,944 B2	7/2009	Sakamoto et al.	9,623,912 B2	4/2017	Schlangen
7,565,945 B2	7/2009	Okada et al.	9,776,481 B2	10/2017	Deckard et al.
7,568,732 B2	8/2009	Schlup, Jr.	9,845,004 B2	12/2017	Hedlund et al.
7,578,523 B2	8/2009	Kosuge et al.	9,873,316 B2	1/2018	Lovold et al.
7,600,603 B2	10/2009	Okada et al.	9,876,316 B2	1/2018	Sugiura et al.
7,604,084 B2	10/2009	Okada et al.	9,895,946 B2	2/2018	Schlangen et al.
7,610,132 B2	10/2009	Yanai et al.	9,988,083 B1	6/2018	Smith
7,632,327 B2	12/2009	Yoo	10,077,087 B2	9/2018	Lubben
7,665,792 B1	2/2010	Flynn	10,300,786 B2	5/2019	Nugteren et al.
			10,358,187 B2	7/2019	Vistad et al.
			10,744,833 B2 *	8/2020	Gell B60G 3/20
			10,746,082 B2	8/2020	Fang et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

10,793,181	B2	10/2020	Bennett et al.	2006/0196722	A1	9/2006	Makabe et al.
10,981,448	B2	4/2021	Safranski et al.	2006/0201270	A1	9/2006	Kobayashi
2002/0093168	A1	7/2002	Hibbert	2006/0207823	A1	9/2006	Okada et al.
2002/0107101	A1	8/2002	Bowen et al.	2006/0207824	A1	9/2006	Saito et al.
2002/0147072	A1	10/2002	Goodell et al.	2006/0207825	A1	9/2006	Okada et al.
2002/0178968	A1	12/2002	Christensen	2006/0208564	A1	9/2006	Yuda et al.
2002/0179356	A1	12/2002	Morii et al.	2006/0212200	A1	9/2006	Yanai et al.
2003/0001409	A1	1/2003	Semple et al.	2006/0219452	A1	10/2006	Okada et al.
2003/0006081	A1	1/2003	Burke et al.	2006/0219469	A1	10/2006	Okada et al.
2003/0089166	A1	5/2003	Mizuno et al.	2006/0219470	A1	10/2006	Imagawa et al.
2003/0172907	A1	9/2003	Nytmot et al.	2006/0273566	A1	12/2006	Hepner et al.
2003/0211913	A1	11/2003	Spitale	2006/0288800	A1	12/2006	Mukai et al.
2004/0026948	A1	2/2004	Novajovsky	2007/0017480	A1	1/2007	Kondo
2004/0079561	A1	4/2004	Ozawa et al.	2007/0028877	A1	2/2007	McDonald et al.
2004/0089492	A1	5/2004	Eide	2007/0068726	A1	3/2007	Shimizu
2004/0099470	A1	5/2004	Tanigaki et al.	2007/0074588	A1	4/2007	Harata et al.
2004/0108159	A1	6/2004	Rondeau et al.	2007/0074589	A1	4/2007	Harata et al.
2004/0153782	A1	8/2004	Fukui et al.	2007/0074927	A1	4/2007	Okada et al.
2004/0159483	A1	8/2004	Imamura et al.	2007/0074928	A1	4/2007	Okada et al.
2004/0187826	A1	9/2004	Kino et al.	2007/0087890	A1	4/2007	Hamrin et al.
2004/0195797	A1	10/2004	Nash et al.	2007/0090621	A1	4/2007	Vigen
2004/0216945	A1	11/2004	Inui	2007/0095601	A1	5/2007	Okada et al.
2004/0221669	A1	11/2004	Shimizu et al.	2007/0096449	A1	5/2007	Okada et al.
2004/0226384	A1	11/2004	Shimizu et al.	2007/0175686	A1	8/2007	Yoshihara
2004/0242090	A1	12/2004	Gibbs	2007/0175696	A1	8/2007	Saito et al.
2004/0262064	A1	12/2004	Lefort	2007/0199753	A1	8/2007	Giese et al.
2005/0012421	A1	1/2005	Fukuda et al.	2007/0225105	A1	9/2007	Bowen
2005/0016784	A1	1/2005	Fecteau	2007/0227810	A1	10/2007	Sakurai et al.
2005/0040000	A1	2/2005	Kelley et al.	2007/0242398	A1	10/2007	Ogawa
2005/0045414	A1	3/2005	Takagi et al.	2007/0246283	A1	10/2007	Giese et al.
2005/0067220	A1	3/2005	Keller et al.	2007/0290493	A1	12/2007	David
2005/0072613	A1	4/2005	Maltais et al.	2008/0023240	A1	1/2008	Sunsdahl et al.
2005/0073140	A1	4/2005	Boucher	2008/0023249	A1	1/2008	Sunsdahl et al.
2005/0073177	A1	4/2005	Bendel et al.	2008/0053743	A1	3/2008	Tomita
2005/0161283	A1	7/2005	Emler	2008/0141957	A1	6/2008	Dea et al.
2005/0173180	A1	8/2005	Hypes et al.	2008/0199253	A1	8/2008	Okada et al.
2005/0173940	A1	8/2005	Smith et al.	2008/0257630	A1	10/2008	Takeshima et al.
2005/0199432	A1	9/2005	Abe et al.	2008/0284124	A1	11/2008	Brady et al.
2005/0199433	A1	9/2005	Abe et al.	2009/0078491	A1	3/2009	Tsutsumikoshi et al.
2005/0205320	A1	9/2005	Girouard et al.	2009/0078492	A1	3/2009	Tsutsumikoshi et al.
2005/0225067	A1	10/2005	Nguyen	2009/0152035	A1	6/2009	Okada et al.
2005/0231145	A1	10/2005	Mukai et al.	2009/0152036	A1	6/2009	Okada et al.
2005/0235767	A1	10/2005	Shimizu et al.	2009/0217908	A1	9/2009	Watanabe et al.
2005/0235768	A1	10/2005	Shimizu et al.	2009/0294197	A1	12/2009	Polakowski et al.
2005/0242677	A1	11/2005	Akutsu et al.	2009/0301830	A1	12/2009	Kinsman et al.
2005/0248116	A1	11/2005	Fanson	2009/0308682	A1	12/2009	Ripley et al.
2005/0252592	A1	11/2005	Albright et al.	2010/0012412	A1	1/2010	Deckard et al.
2005/0257989	A1	11/2005	Iwami et al.	2010/0060026	A1	3/2010	Bowers
2005/0257990	A1	11/2005	Shimizu	2010/0071982	A1	3/2010	Giese et al.
2005/0267660	A1	12/2005	Fujiwara et al.	2010/0089355	A1	4/2010	Fredrickson et al.
2005/0279552	A1	12/2005	Schuehmacher et al.	2010/0090430	A1	4/2010	Weston
2006/0022619	A1	2/2006	Koike et al.	2010/0108427	A1	5/2010	Richer
2006/0042862	A1	3/2006	Saito et al.	2010/0133036	A1	6/2010	Okada et al.
2006/0043754	A1	3/2006	Smith	2010/0170231	A1	7/2010	Yamaguchi
2006/0055139	A1	3/2006	Furumi et al.	2010/0269771	A1	10/2010	Wermuth et al.
2006/0065472	A1	3/2006	Ogawa et al.	2011/0088968	A1	4/2011	Kempf et al.
2006/0075840	A1	4/2006	Saito et al.	2011/0100340	A1	5/2011	Mukkala et al.
2006/0076180	A1	4/2006	Saito et al.	2011/0139529	A1	6/2011	Eichenberger et al.
2006/0085966	A1	4/2006	Kerner et al.	2011/0240395	A1	10/2011	Hurd et al.
2006/0108174	A1	5/2006	Saito et al.	2011/0298189	A1	12/2011	Schneider et al.
2006/0162990	A1	7/2006	Saito et al.	2012/0143465	A1	6/2012	Kim
2006/0169510	A1	8/2006	Visscher	2012/0160593	A1	6/2012	Nozaki et al.
2006/0169514	A1	8/2006	Lim et al.	2012/0205902	A1	8/2012	Beavis et al.
2006/0169525	A1	8/2006	Saito et al.	2012/0223500	A1	9/2012	Kinsman et al.
2006/0175124	A1	8/2006	Saito et al.	2012/0261210	A1	10/2012	Kondou
2006/0180370	A1	8/2006	Polakowski et al.	2013/0048409	A1	2/2013	Fujii
2006/0180385	A1	8/2006	Yanai et al.	2013/0125535	A1	5/2013	Schultz
2006/0185741	A1	8/2006	McKee	2013/0175779	A1	7/2013	Kvien et al.
2006/0185919	A1	8/2006	Hibbert et al.	2013/0180798	A1	7/2013	Matsushima et al.
2006/0185927	A1	8/2006	Sakamoto et al.	2013/0233265	A1	9/2013	Zurface et al.
2006/0191734	A1	8/2006	Kobayashi	2013/0240272	A1	9/2013	Gass et al.
2006/0191735	A1	8/2006	Kobayashi	2013/0256050	A1	10/2013	Novotny et al.
2006/0191737	A1	8/2006	Kobayashi	2014/0124279	A1	5/2014	Schlangen et al.
2006/0191739	A1	8/2006	Koga	2014/0210234	A1	7/2014	Ricketts et al.
2006/0196721	A1	9/2006	Saito et al.	2014/0262583	A1	9/2014	Url
				2014/0353956	A1	12/2014	Bjerketvedt et al.
				2015/0165886	A1	6/2015	Bennett et al.
				2015/0197128	A1	7/2015	Miller et al.
				2015/0210115	A1	7/2015	David et al.

(56)

References Cited**U.S. PATENT DOCUMENTS**

2015/0210319	A1	7/2015	Tiramani	
2015/0354149	A1	12/2015	Dahm et al.	
2016/0347137	A1	12/2016	Despres-Nadeau et al.	
2016/0347350	A1	12/2016	Heon et al.	
2017/0101142	A1	4/2017	Hedlund et al.	
2017/0137059	A1	5/2017	Ohba et al.	
2017/0210192	A1	7/2017	Reybrouck et al.	
2017/0274935	A1	9/2017	Deckard et al.	
2018/0154717	A1	6/2018	Neu et al.	
2018/0170134	A1	6/2018	Schlangen et al.	
2018/0170136	A1*	6/2018	Gell	B60G 3/20
2019/0055875	A1	2/2019	Fang et al.	
2019/0248405	A1	8/2019	Bennett et al.	
2021/0016823	A1	1/2021	Otto et al.	
2021/0354542	A1	11/2021	Schleif et al.	

FOREIGN PATENT DOCUMENTS

AU	2017204255	A1	7/2017
CA	1150752	A	7/1983
CA	2265693	A1	3/1998
CA	2561337		9/1999
CH	317335	A	11/1956
CN	101511664	A	8/2009
CN	102056788	A	5/2011
CN	103419869	A	12/2013
CN	104661903	A	5/2015
CN	107089108	A	8/2017
CN	108357598	A	8/2018
CN	109803872	A	5/2019
DE	0116605		2/1900
DE	0351667	C	4/1922
DE	1755101		4/1971
DE	3033707		4/1982
DE	4122690	A1	1/1992
DE	102010017784	A1	1/2012
EP	0238077	A2	9/1987
EP	0287038	A2	10/1988
EP	0709247	A2	5/1996
EP	0794096	A2	9/1997
EP	1215107	A1	6/2002
EP	1557345	A2	7/2005
EP	1564123	A2	8/2005
EP	1769998	A1	4/2007
EP	3328717	A1	6/2018
FR	2678886	A1	1/1993
JP	62-218209	A	9/1987
JP	01-113527	A	5/1989
JP	11-334447	A	12/1999
JP	2000-177434	A	6/2000
JP	2001-065344	A	3/2001
JP	2003-011843	A	1/2003
JP	3385382	B2	3/2003
JP	2003-343373	A	12/2003
JP	2005-193788	A	7/2005
JP	2006-232058	A	9/2006
JP	2006-232061	A	9/2006
JP	2006-256579	A	9/2006
JP	2006-256580	A	9/2006
JP	2006-281839	A	10/2006
JP	2007-038709	A	2/2007
JP	2007-106319	A	4/2007
JP	2018-523603	A	8/2018
WO	95/01503	A1	1/1995
WO	97/46407	A1	12/1997
WO	98/12095	A2	3/1998
WO	01/05613	A1	1/2001
WO	02/87957	A1	11/2002
WO	2007/100751	A2	9/2007
WO	2009/062823	A1	5/2009
WO	2009/096998	A1	8/2009
WO	2009/114414	A1	9/2009
WO	2009/151995	A1	12/2009
WO	2011/093847	A1	8/2011
WO	2011/099959	A1	8/2011

WO	2013/119958	A2	8/2013
WO	2014/047488	A1	3/2014
WO	2015/071904	A1	5/2015
WO	2015/105974	A2	7/2015
WO	2015/105976	A1	7/2015
WO	2015/184354	A1	12/2015
WO	2017/023726	A1	2/2017

OTHER PUBLICATIONS

Brochure, Fully Equipped LTATV Sales, available at <http://chrishainesoffroad.com/vehicles/pdf>, last accessed Jul. 15, 2014, copyright 2010-2011; 2 pages.

Brochure, Strike Razer, Advanced Mobile System for Rapid Response; 5 pages.

Communication Pursuant to Article 94(3) EPC issued by the European Patent Office, dated Nov. 14, 2018, for European Patent Application No. 15700942.4; 4 pages.

Communication Pursuant to Article 94(3) of the European Patent Convention, dated Feb. 6, 2018, for European Patent Application No. 15700942.4; 4 pages.

Decision on Appeal, Inter Partes Reexamination Control U.S. Appl. No. 95/002,010 of U.S. Pat. No. 7,854,285, Appeal 2015-004514, Sep. 28, 2015; 32 pages.

Decision on Appeal, Inter Partes Reexamination Control U.S. Appl. No. 95/002,011 of U.S. Pat. No. 8,151,923, Appeal 2015-004515, Sep. 28, 2015; 36 pages.

Examination Report No. 1 issued by the Australian Government IP Australia, dated Jul. 24, 2017, for Australian Patent Application No. 2015266694; 4 pages.

Examination Report No. 1 issued by the Australian Government IP Australia, dated Jun. 28, 2018, for Australian Patent Application No. 2017204255; 3 pages.

Examination Report No. 1 issued by the Australian Government IP Australia, dated Nov. 19, 2018, for Australian Patent Application No. 2016303426; 3 pages.

Examination Report No. 1 issued by the Australian Government IP Australia, dated Sep. 29, 2016, for Australian Patent Application No. 2013317853; 5 pages.

International Preliminary Report on Patentability issued by the European Patent Office, dated Dec. 13, 2010, for International Patent Application No. PCT/US2009/045900; 7 pages.

International Preliminary Report on Patentability issued by the European Patent Office, dated Jul. 12, 2016, for International Patent Application No. PCT/US2015/010621; 12 pages.

International Preliminary Report on Patentability issued by the European Patent Office, dated Jun. 10, 2008, for International Patent Application No. PCT/US2007/004895; 22 pages.

International Preliminary Report on Patentability issued by the European Patent Office, dated Jun. 17, 2016, for International Patent Application No. PCT/US2015/010623; 20 pages.

International Preliminary Report on Patentability issued by the European Patent Office, dated Mar. 30, 2011, for International Patent Application No. PCT/US2009/066110; 12 pages.

International Preliminary Report on Patentability issued by the European Patent Office, mailed Jun. 4, 2014, for International Patent Application No. PCT/US2013/025354; 18 pages.

International Preliminary Report on Patentability issued by the International Bureau of WIPO, dated Jul. 19, 2011, for International Patent Application No. PCT/US2009/066093; 9 pages.

International Preliminary Report on Patentability received for PCT Patent Application No. PCT/US2019/017438, mailed on Jun. 18, 2020, 17 pages.

International Search Report and Written Opinion issued by the European Patent Office, dated Aug. 8, 2015, for International Patent Application No. PCT/US2015/033327; 8 pages.

International Search Report and Written Opinion issued by the European Patent Office, dated Jan. 16, 2014, for International Patent Application No. PCT/US2013/061002; 13 pages.

International Search Report and Written Opinion issued by the European Patent Office, dated Nov. 6, 2007, for International Patent Application No. PCT/US2007/004895; 20 pages.

International Search Report and Written Opinion issued by the

(56)

References Cited

OTHER PUBLICATIONS

European Patent Office, dated Sep. 18, 2014, for International Patent Application No. PCT/US2013/025354; 13 pages.
 International Search Report and Written Opinion issued by the European Patent Office, mailed Aug. 12, 2014, for International Patent Application No. PCT/US2015/010621; 17 pages.
 International Search Report and Written Opinion issued by the European Patent Office, mailed Jul. 2, 2010, for International Patent Application No. PCT/US2009/066093; 14 pages.
 International Search Report and Written Opinion issued by the European Patent Office, mailed Mar. 24, 2015, for International Patent Application No. PCT/US2015/010623; 11 pages.
 International Search Report and Written Opinion received for PCT Patent Application No. PCT/US2019/017438, mailed on Jul. 11, 2019, 22 pages.
 International Search Report issued by the European Patent Office, dated Aug. 4, 2009, for International Patent Application No. PCT/US2009/045900; 3 pages.
 International Search Report issued by the European Patent Office, dated Mar. 18, 2010, for International Patent Application No. PCT/US2009/066110; 5 pages.
 Invitation to Pay Additional Fees received for PCT Patent Application No. PCT/US2019/017438, mailed on May 17, 2019, 19 pages.

Strike Razor: Light Tactical Vehicle, Advanced Mobile Systerus, copyright 2011; 2 pages.
 SuperATVMedia: "Kawasaki Teryx 4 6" Lift Kit-SuperATV", Dec. 3, 2015, https://www.youtube.com/watch?r=a6ewNg5De_g; 3 pages.
 Unknown: "6" Lift Kit—Kawasaki Teryx4", Nov. 13, 2015, Madison, Indiana Retrieved from the Internet: URL: <https://www.pureoffroad.com/assets/images/superatv/IN-LK-K-TRX4-6.pdf>.
 Unknown: "Kawasaki Teryx 6" Lift Kit", Jan. 23, 2016, Retrieved from the Internet: URL: <https://www.superatv.com/kawasaki-teryx-6-lift-kit>.
 Webpage, RP Advanced Mobile Systems: Military SxS Tactical Defense Vehicles, available at <http://web.archive.org/web/20120324210549/http://www.rpadvancedmobilesystems.com/>, available as early as Mar. 24, 2012, last accessed Jul. 15, 2014; 4 pages.
 Written Opinion issued by the European Patent Office, dated Aug. 27, 2008, for International Patent Application No. PCT/US2008/003485; 7 pages.
 Written Opinion issued by the European Patent Office, dated Oct. 2, 2008, for International Patent Application No. PCT/US2008/003483; 10 pages.
 Written Opinion issued by the European Patent Office, dated Sep. 1, 2008, for International Patent Application No. PCT/US2008/003480; 12 pages.
 Written Opinion of the International Searching Authority, dated Aug. 4, 2009, for International Patent Application No. PCT/US2009/045900; 6 pages.

* cited by examiner

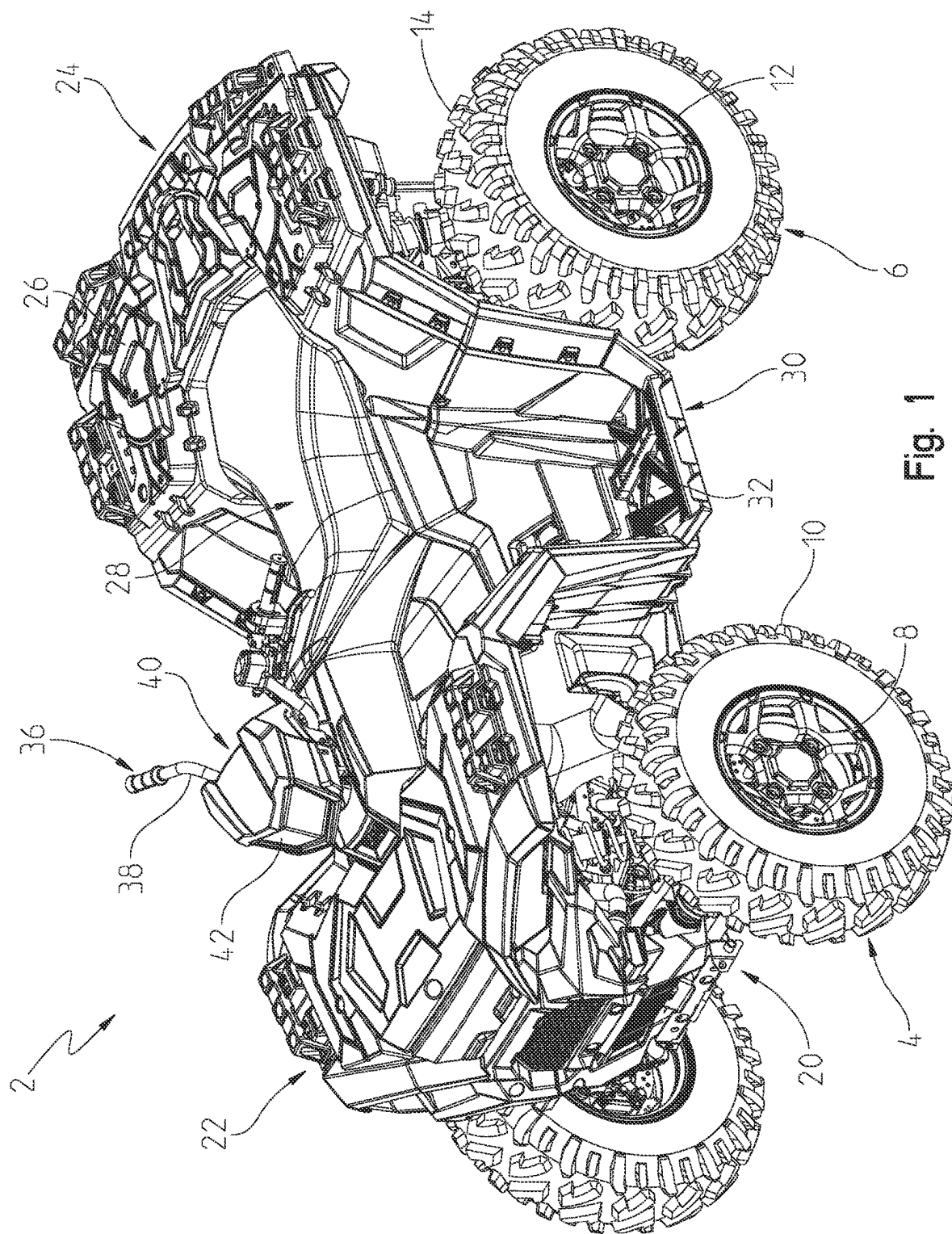
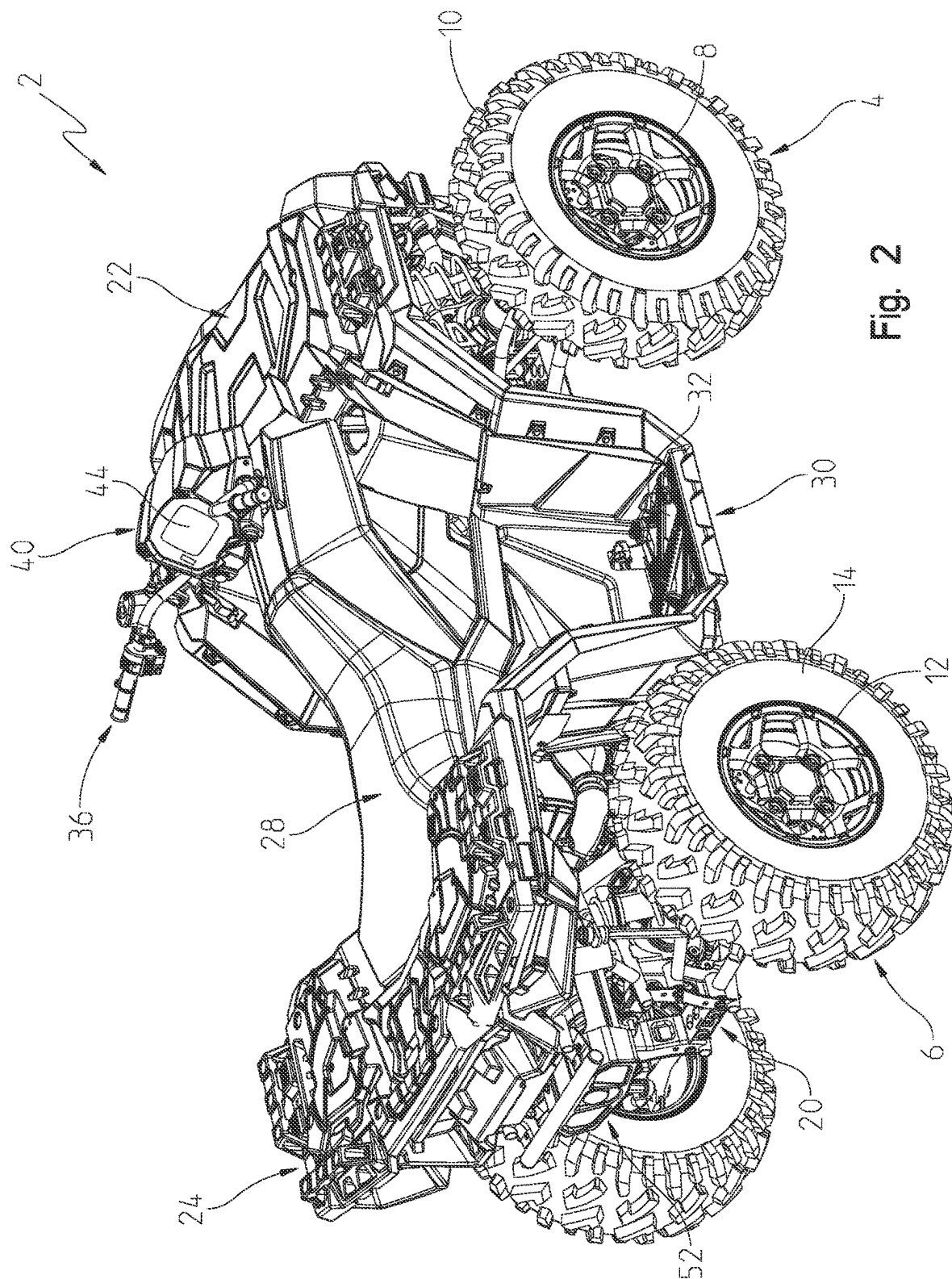


Fig. 1



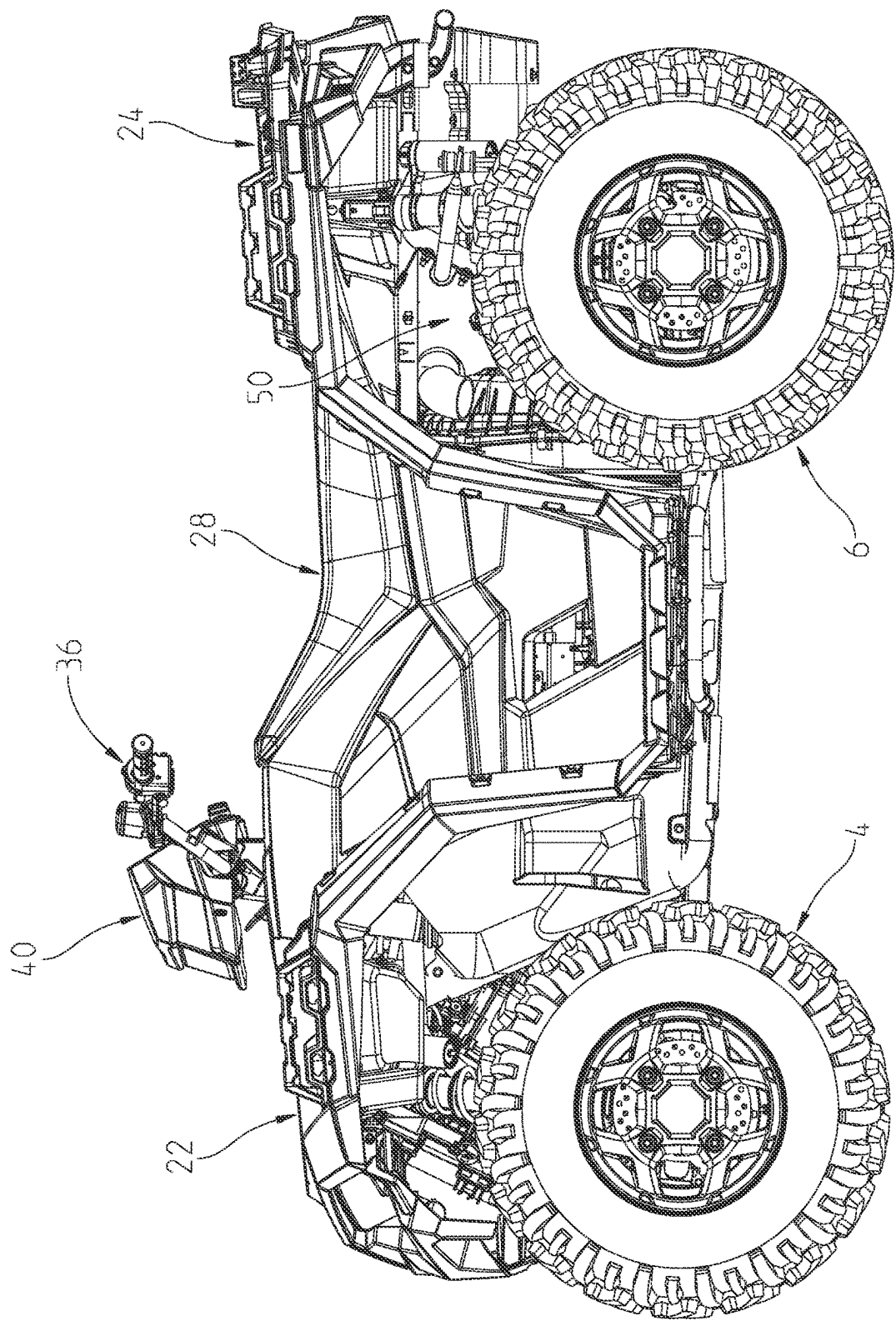


Fig. 3

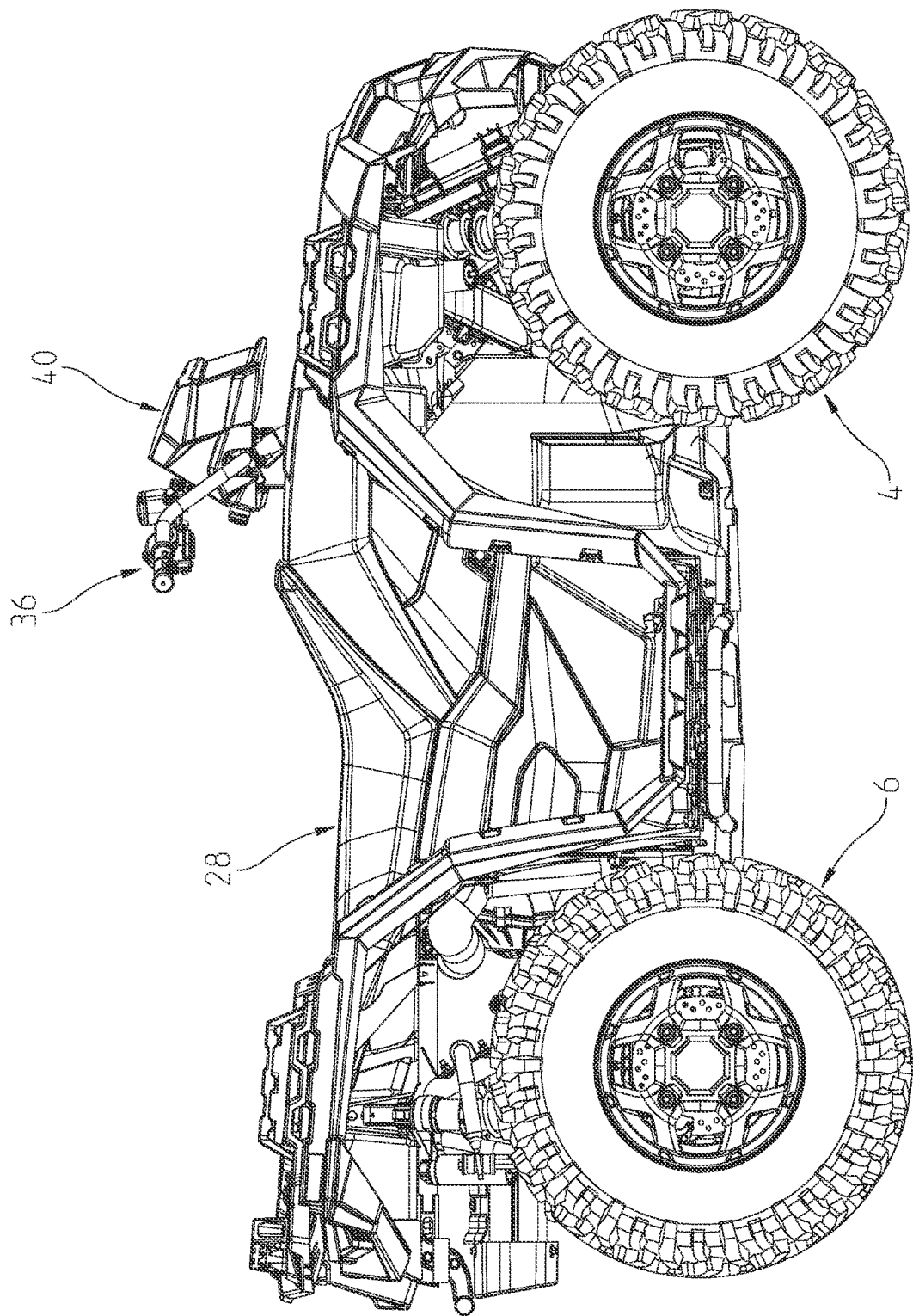


Fig. 4

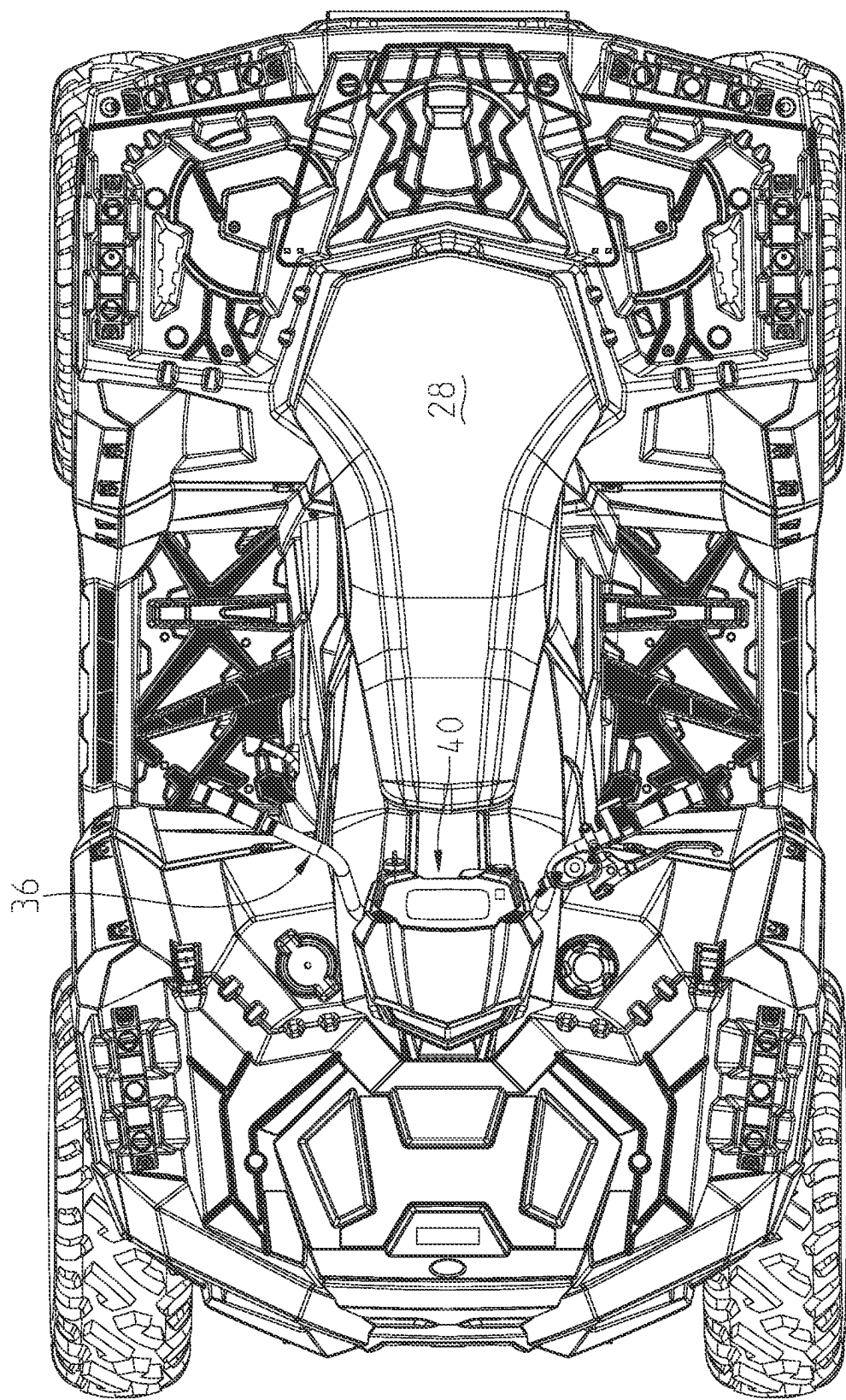


Fig. 5

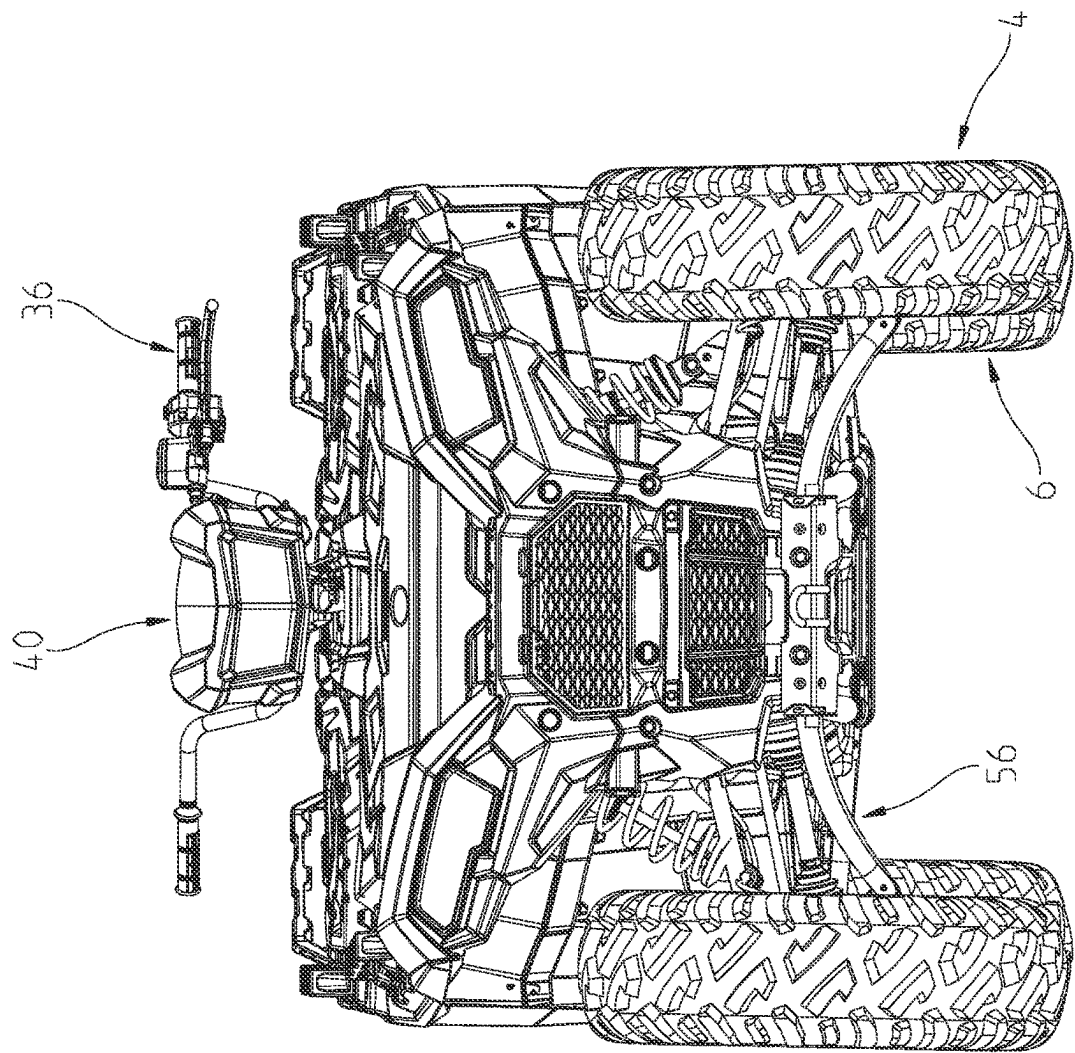


Fig. 6

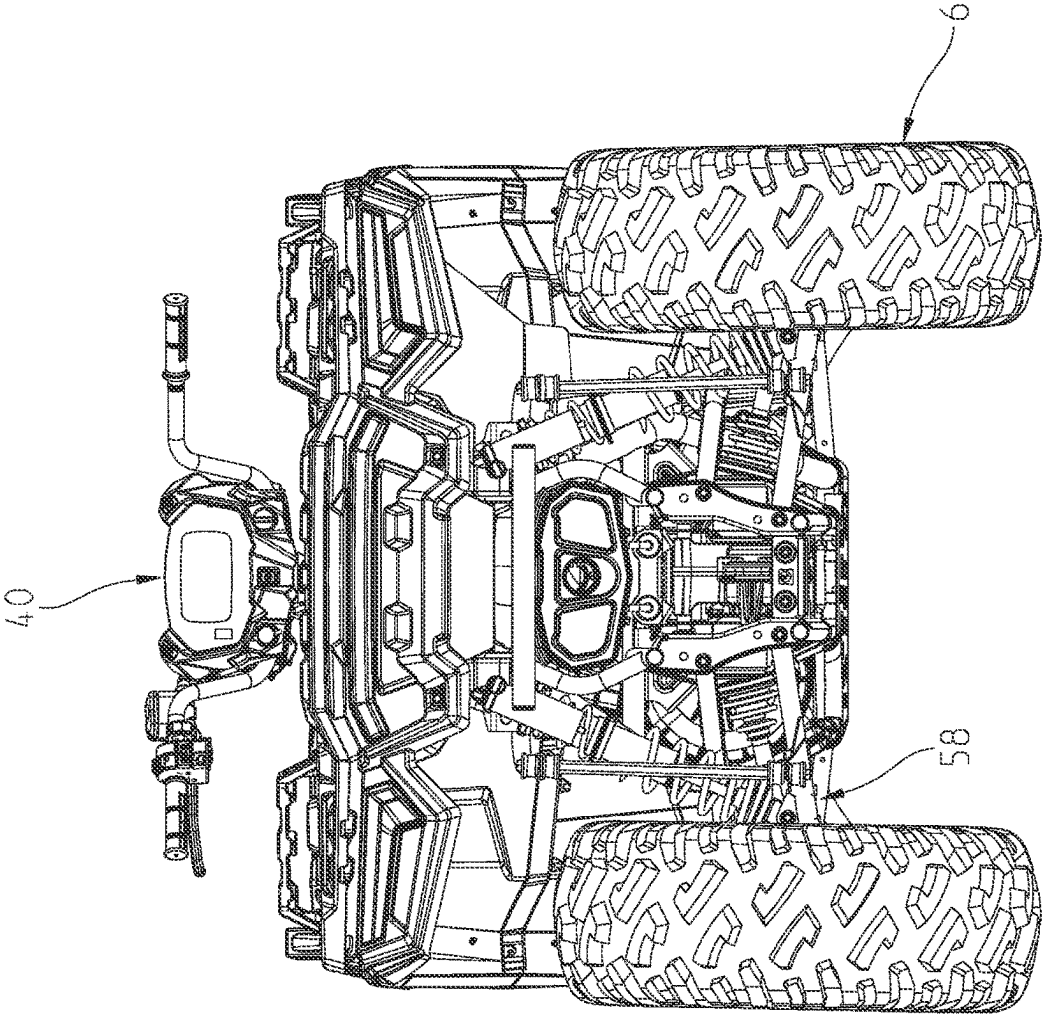


Fig. 7

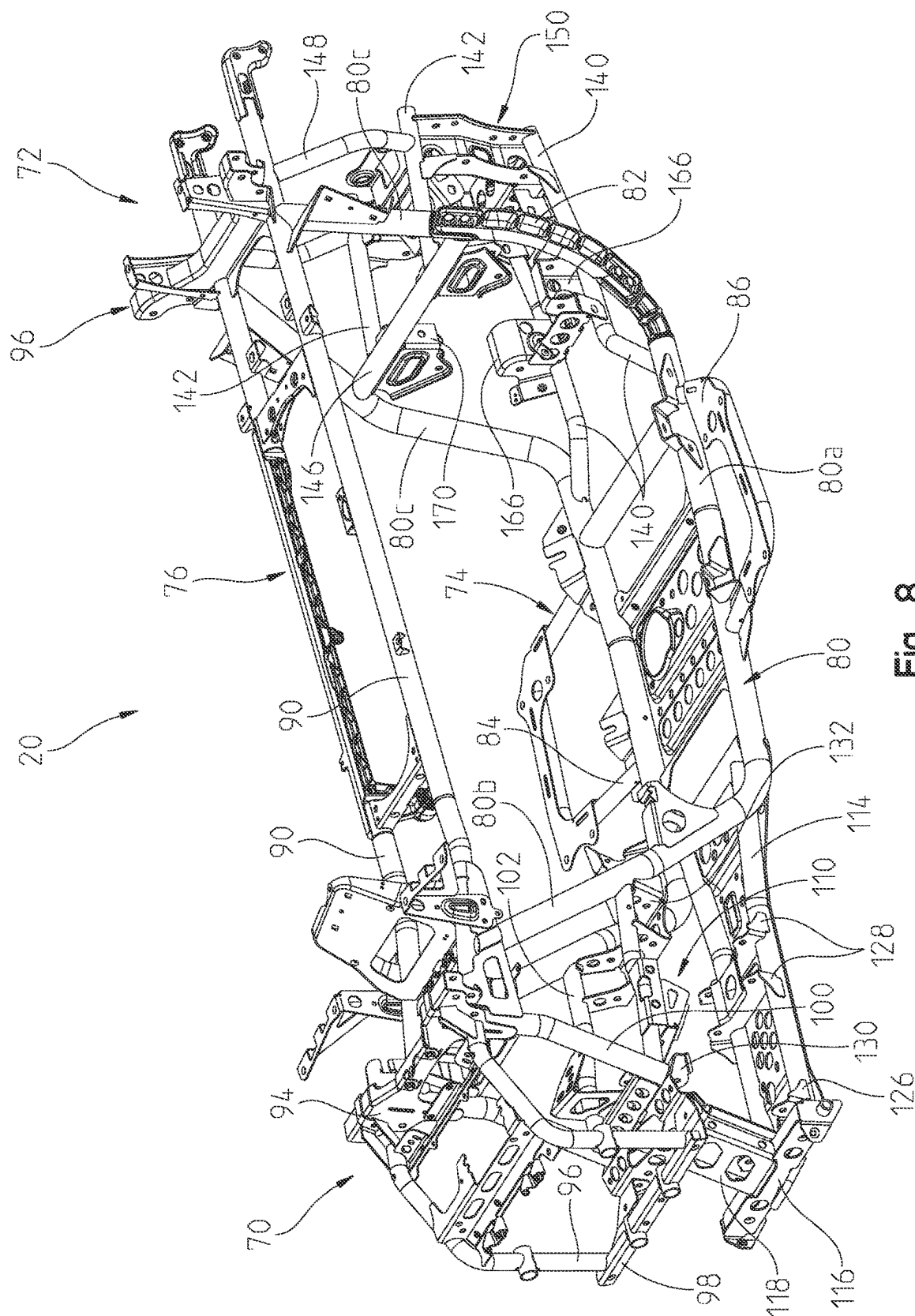
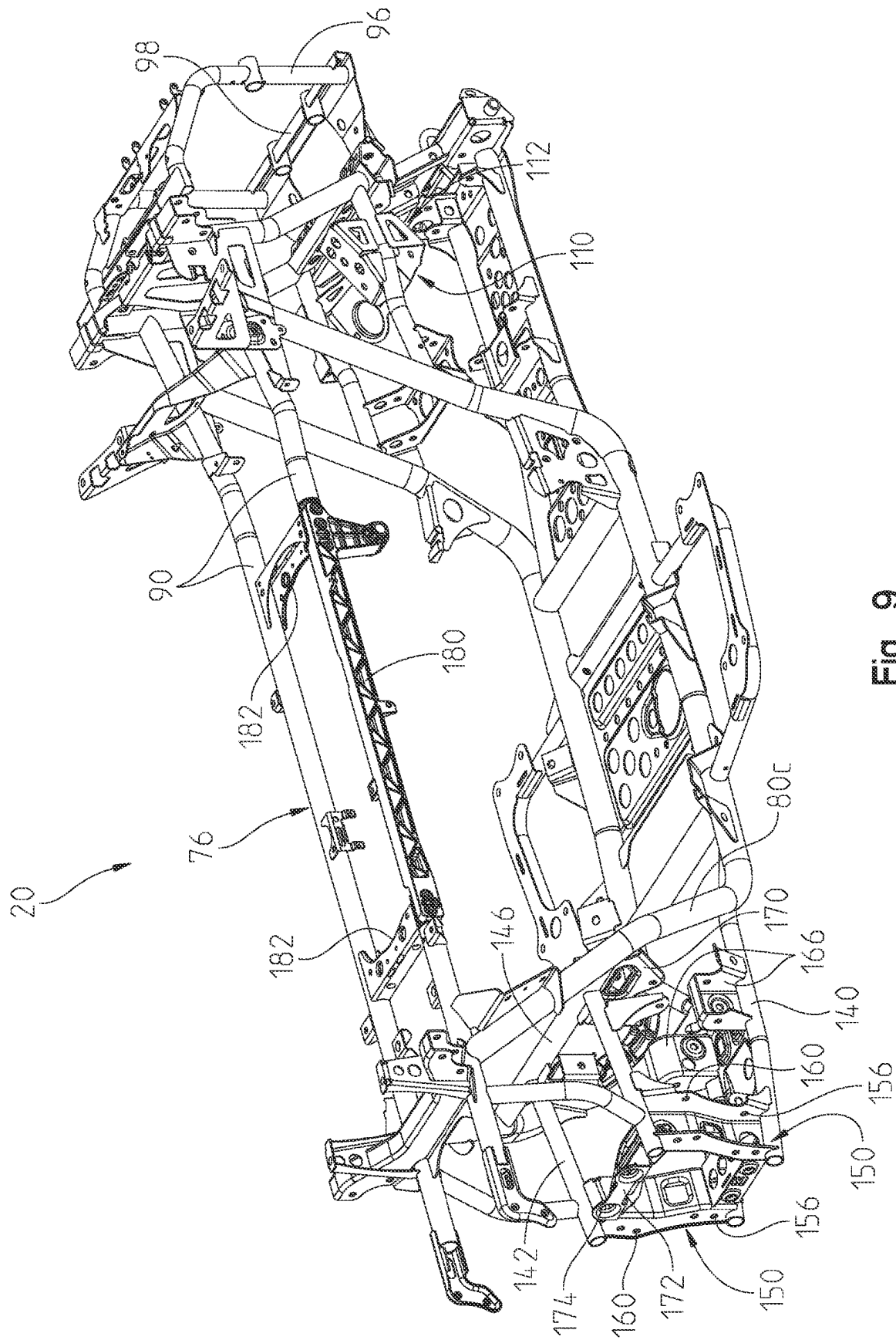


Fig. 8



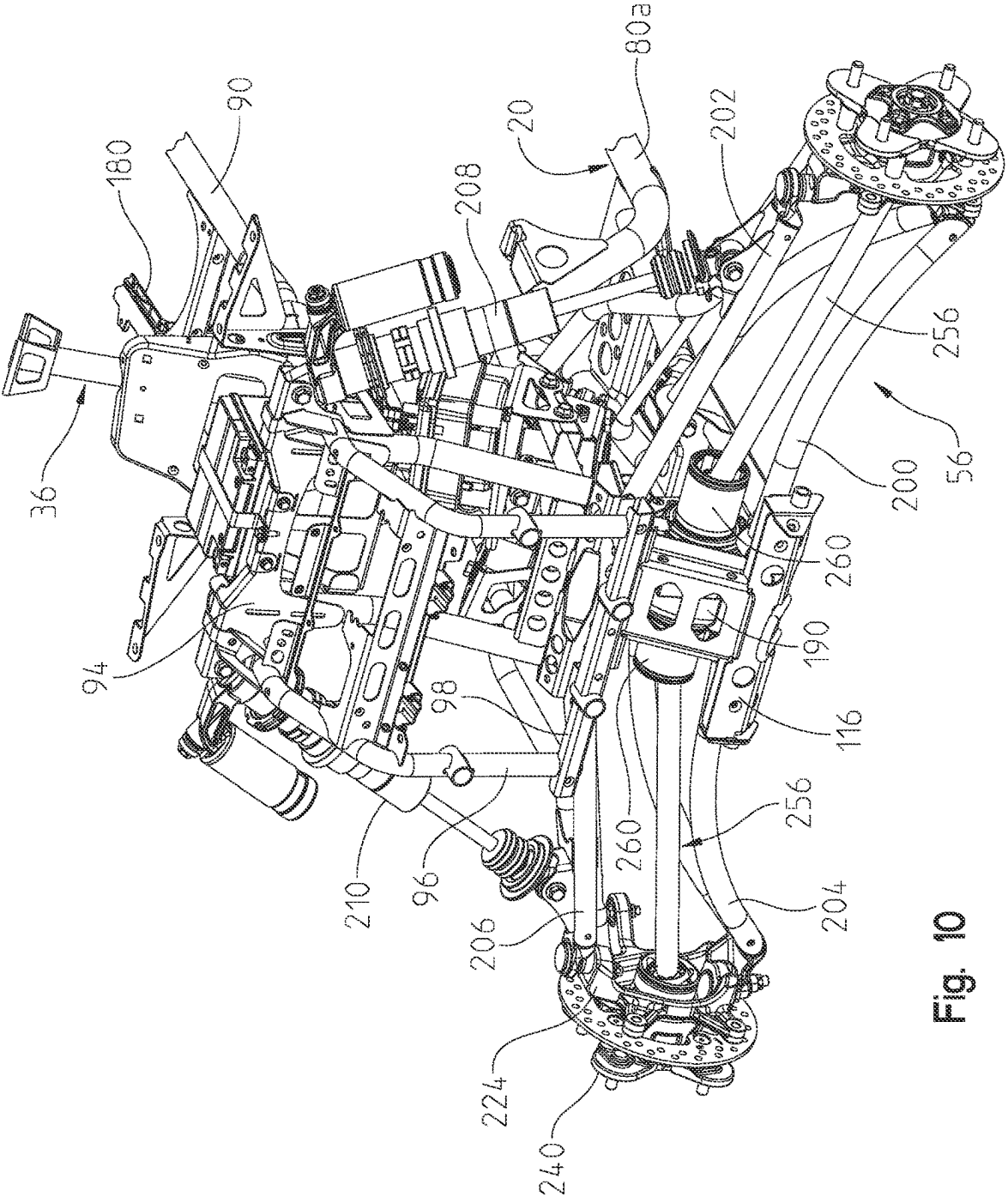
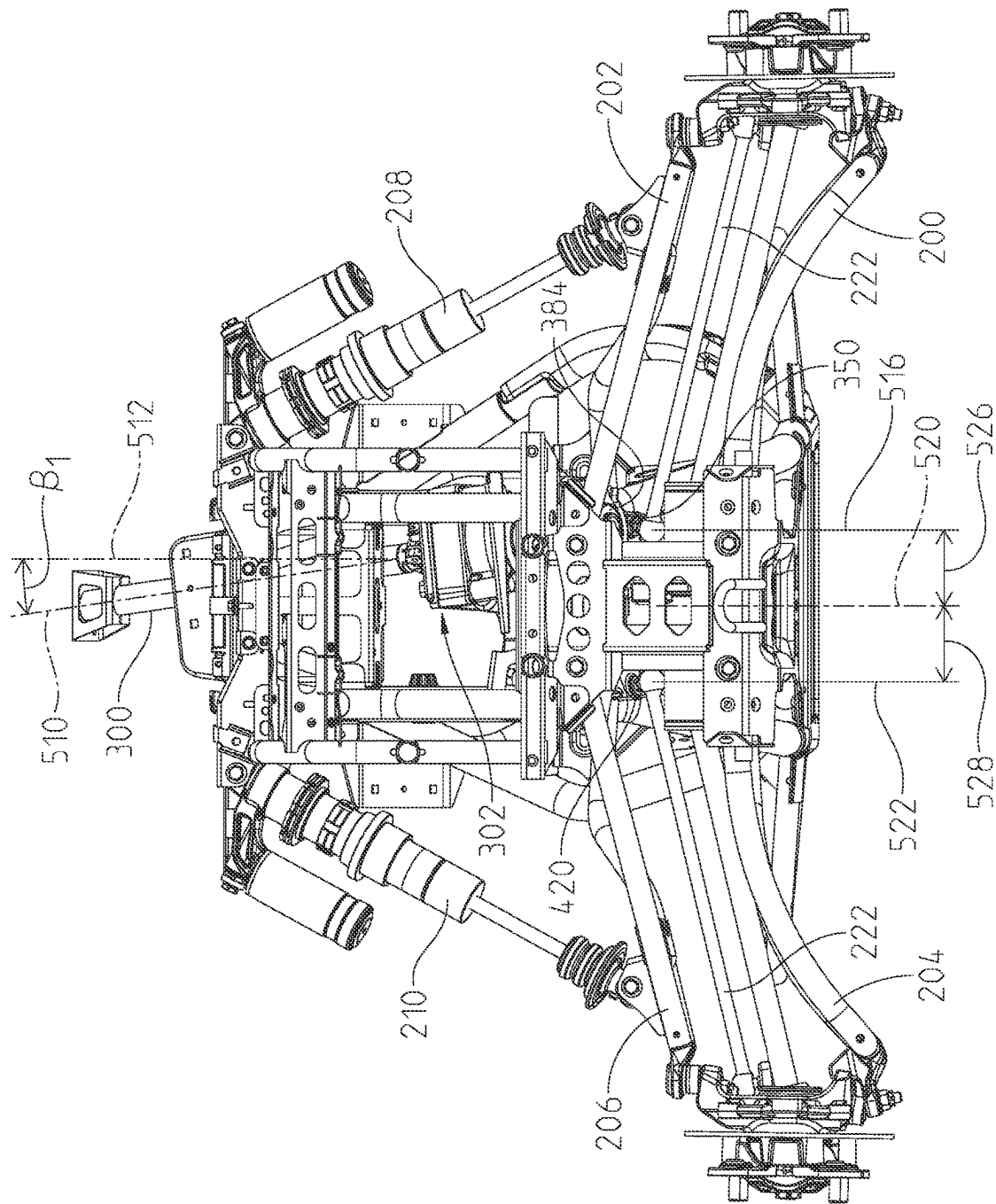


Fig. 10





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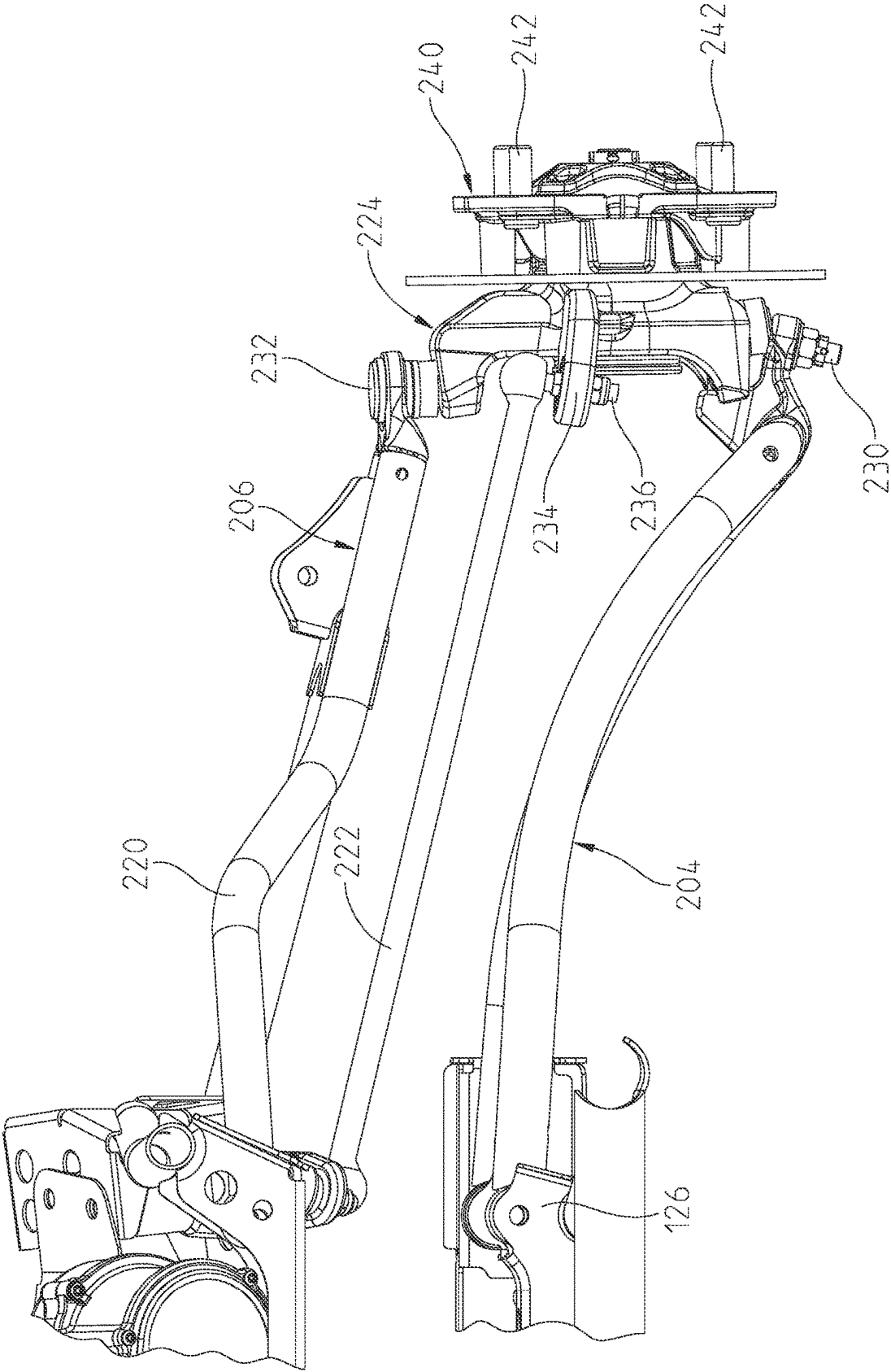


Fig. 12

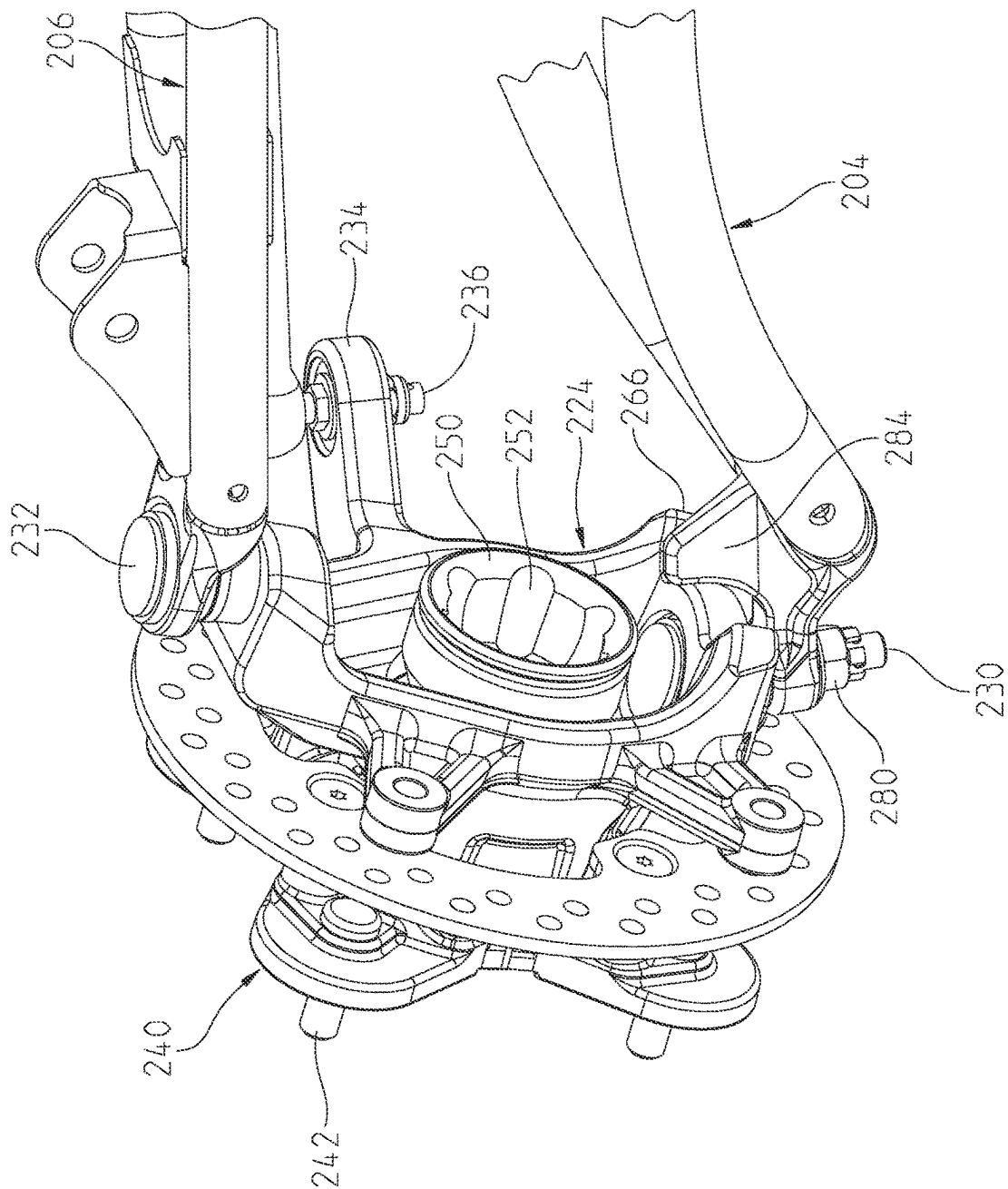


Fig. 13

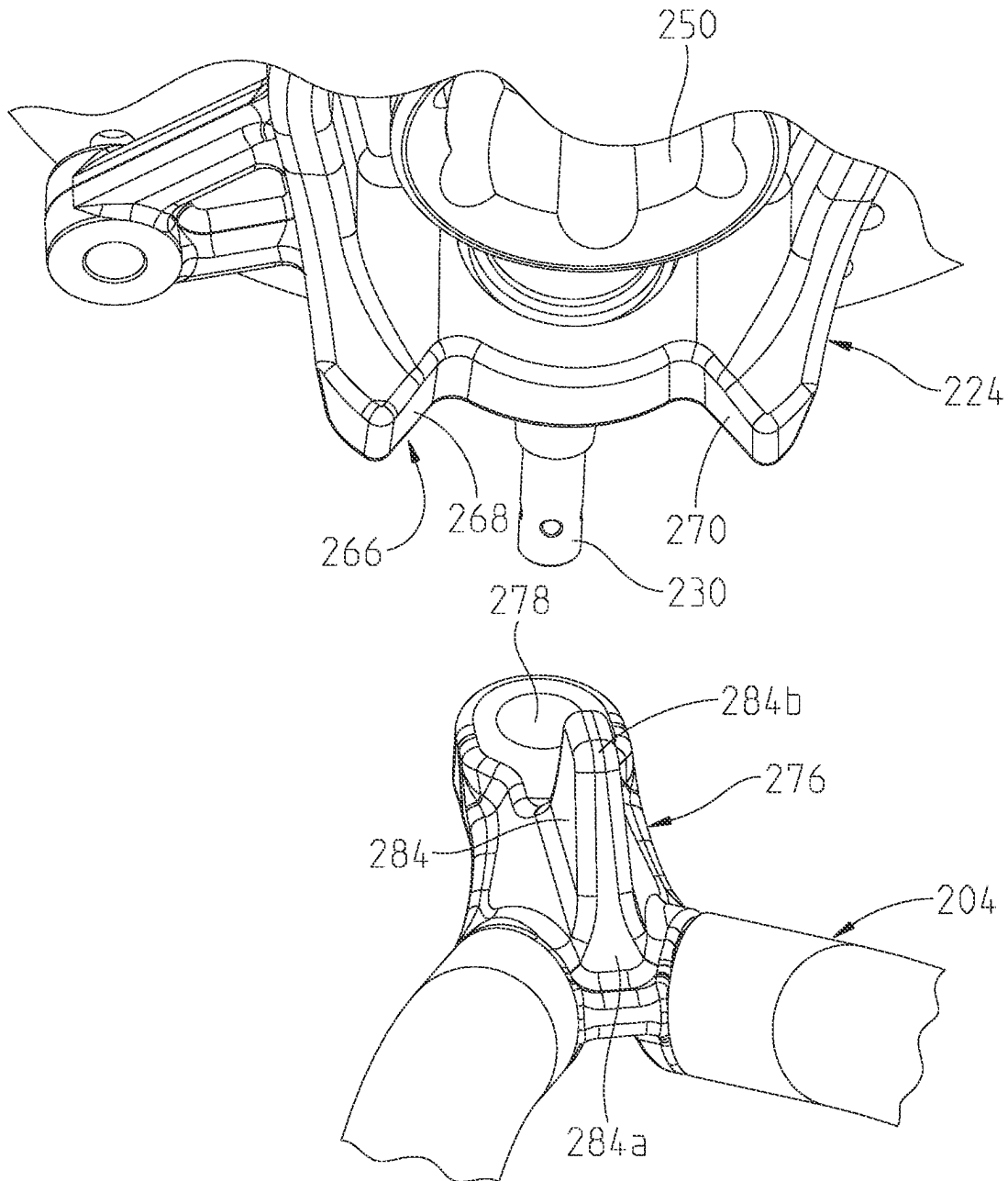


Fig. 14

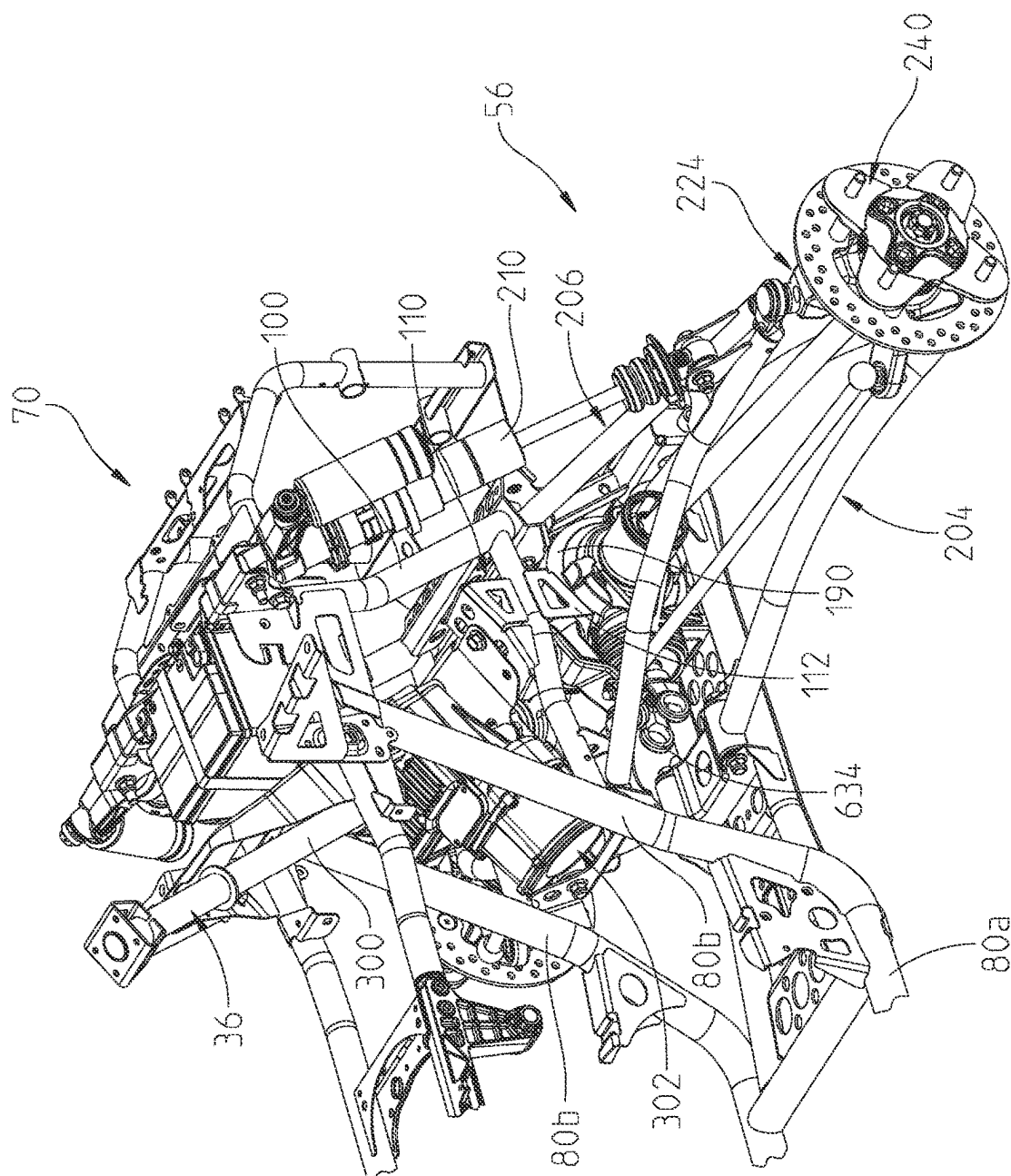


Fig. 15

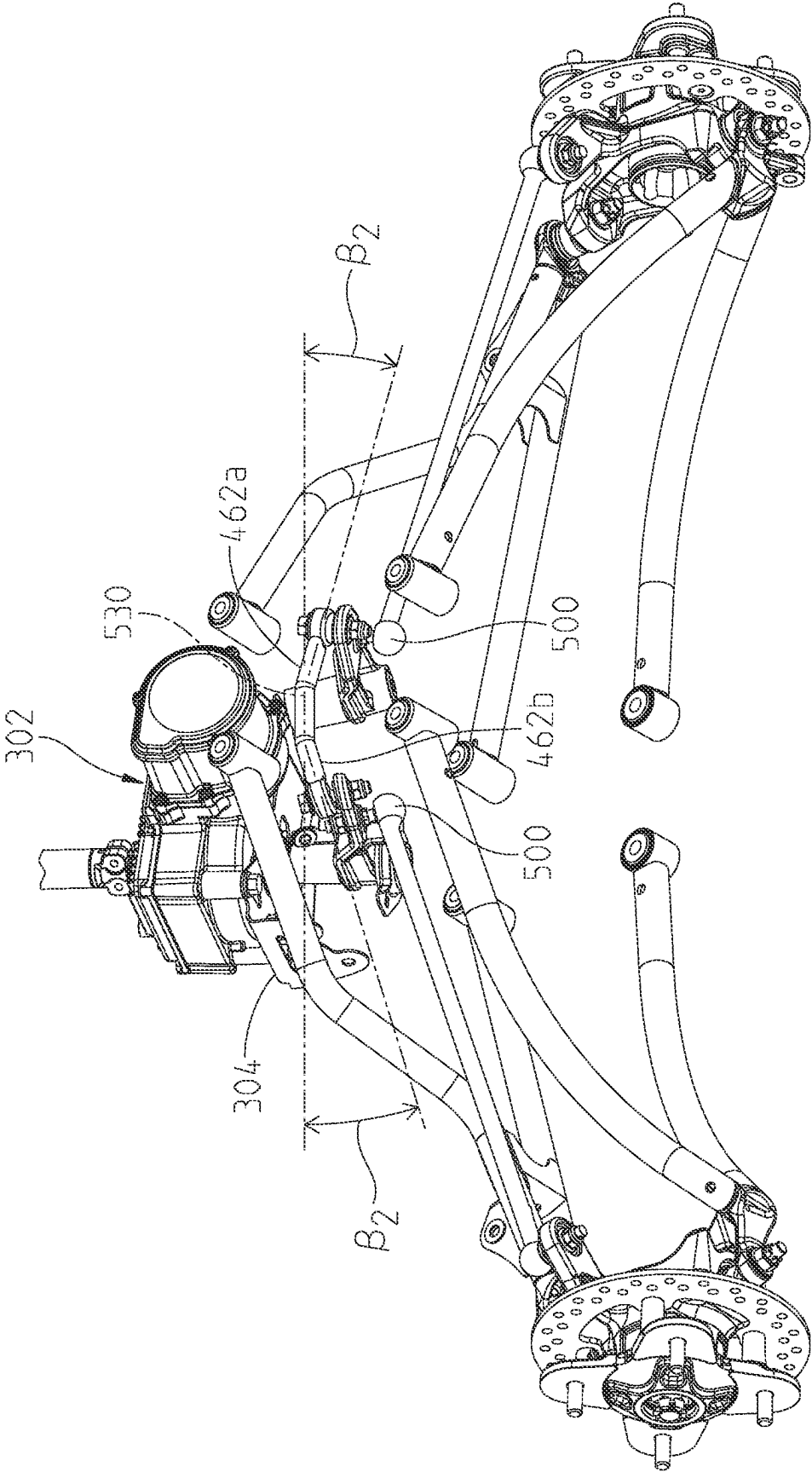


Fig. 16

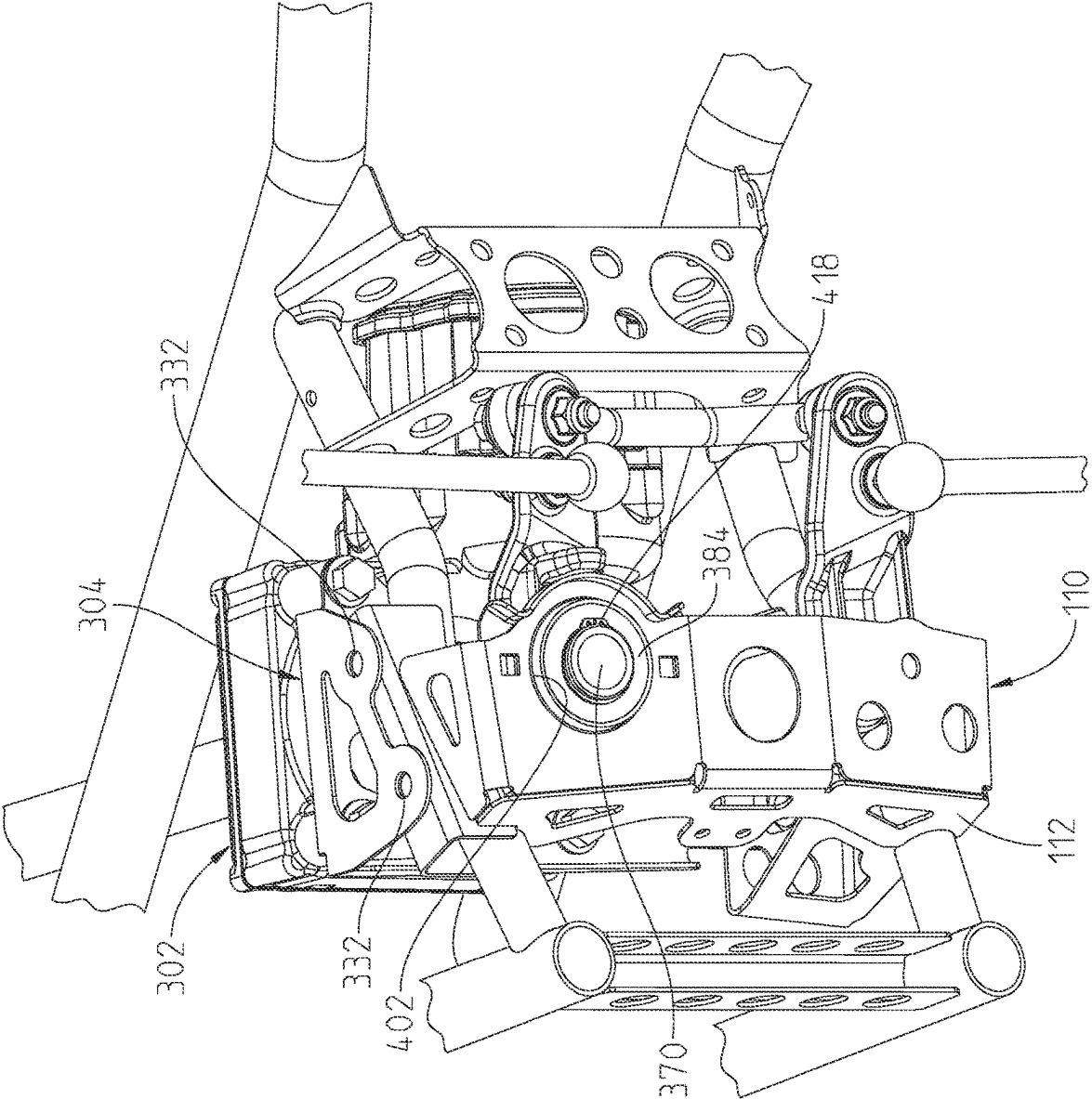


Fig. 17

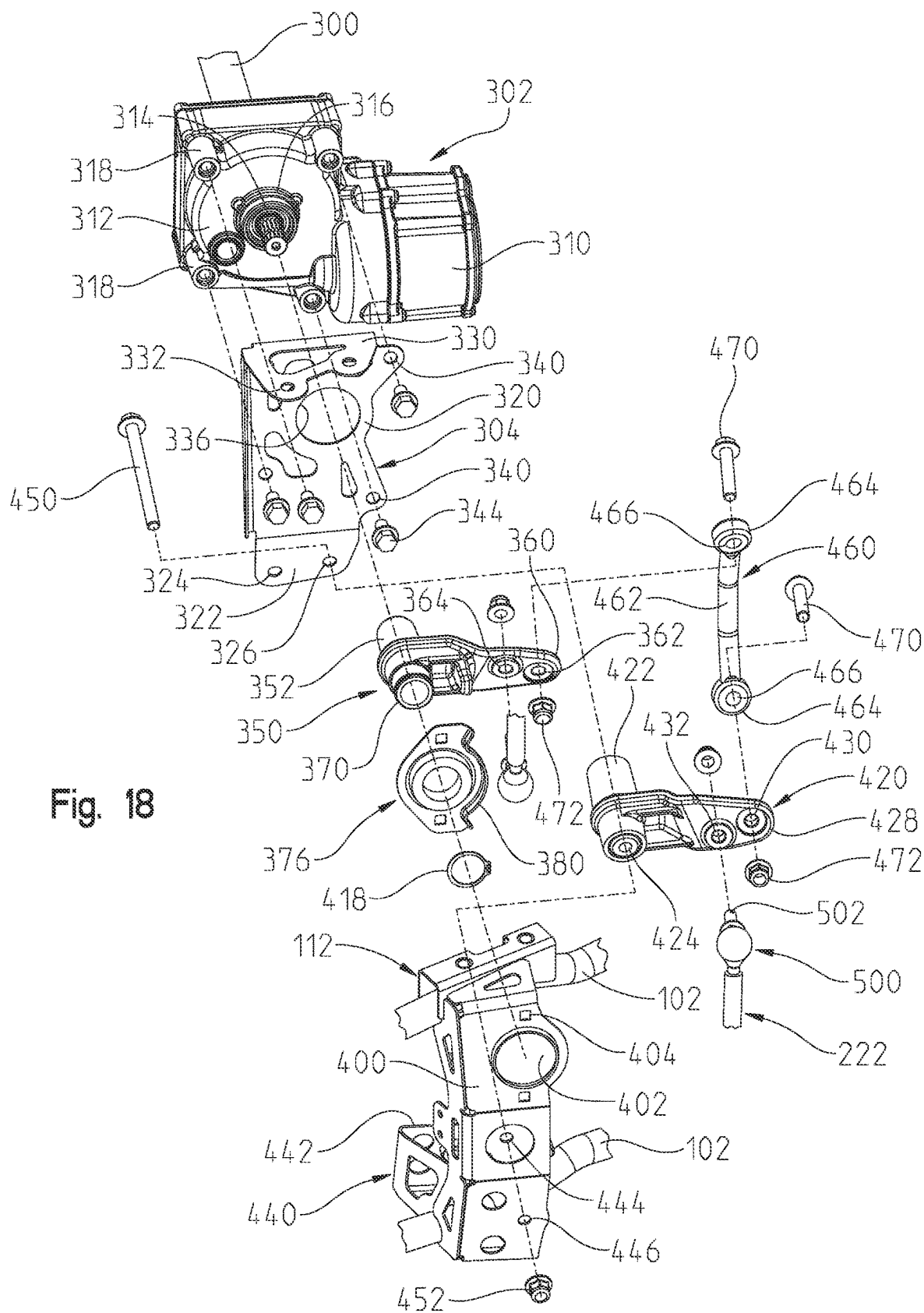
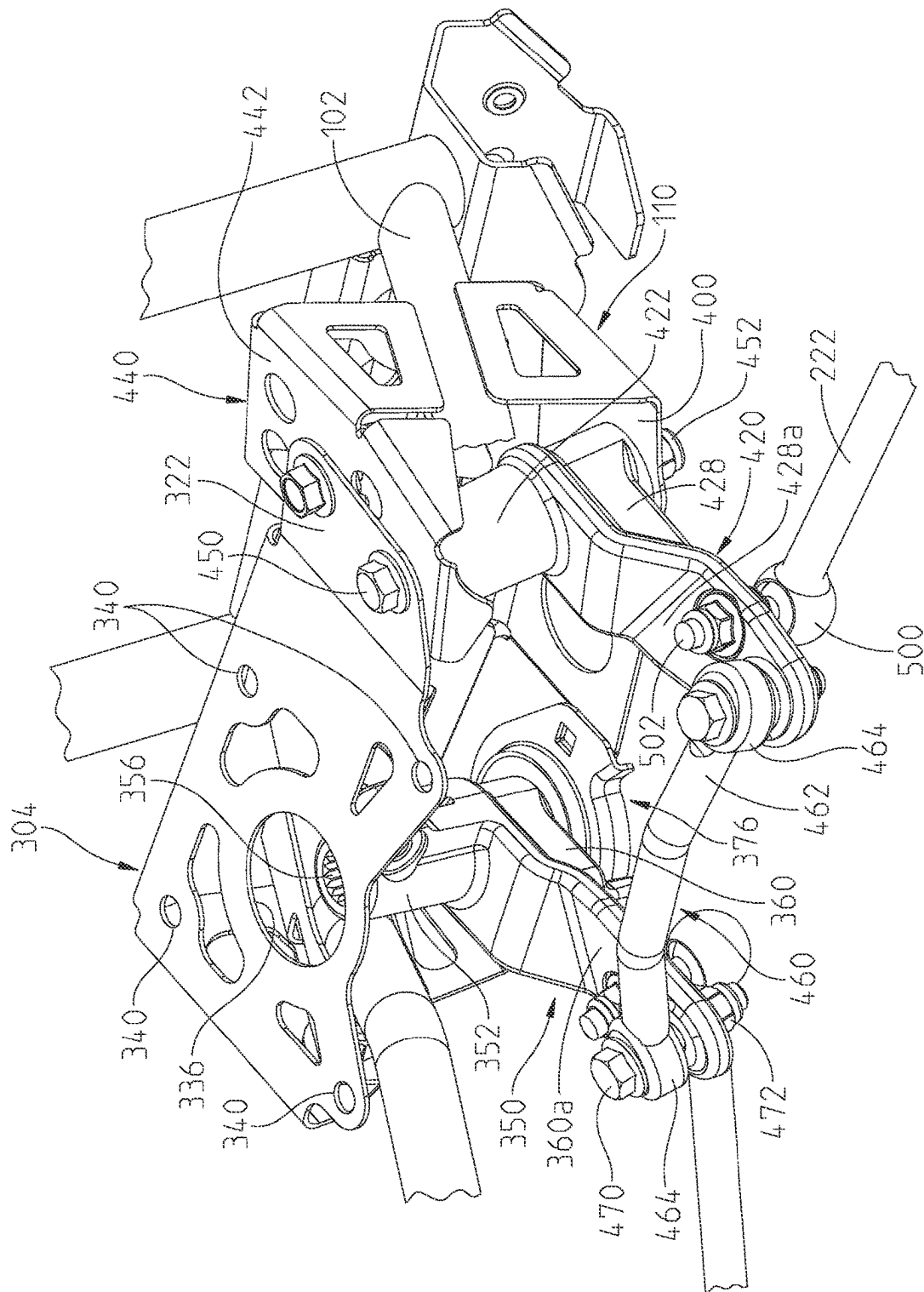


Fig. 18



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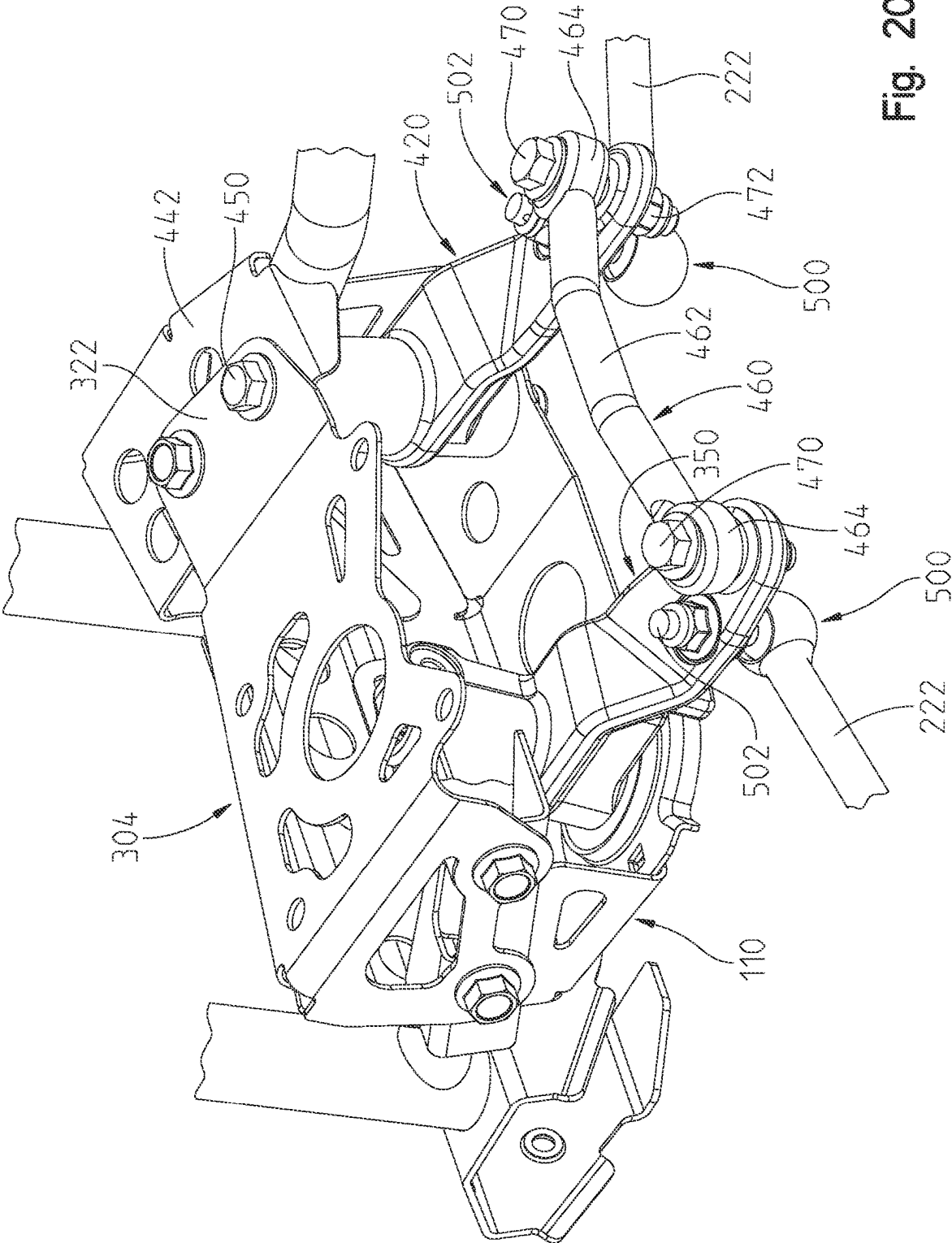


Fig. 20

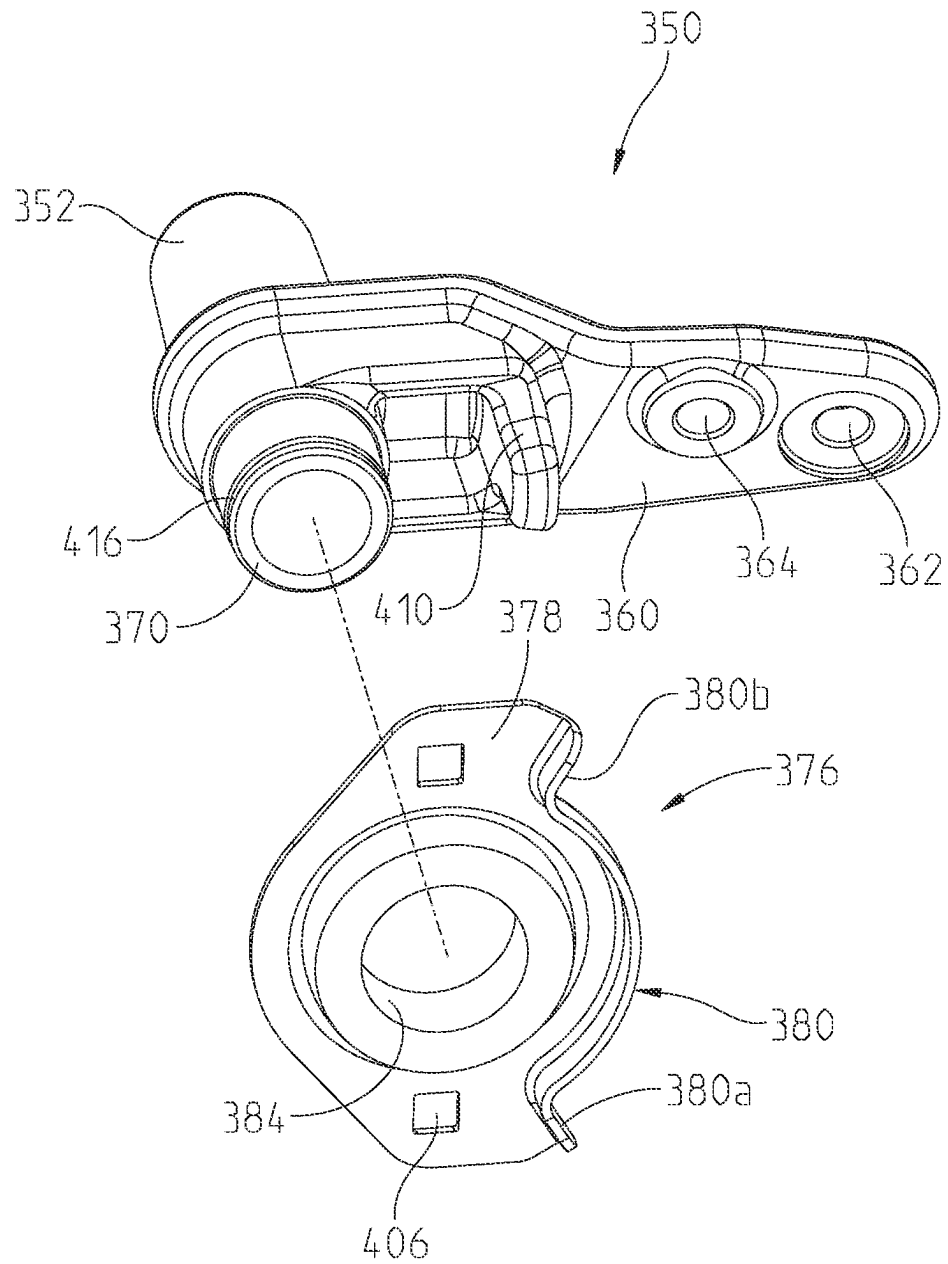
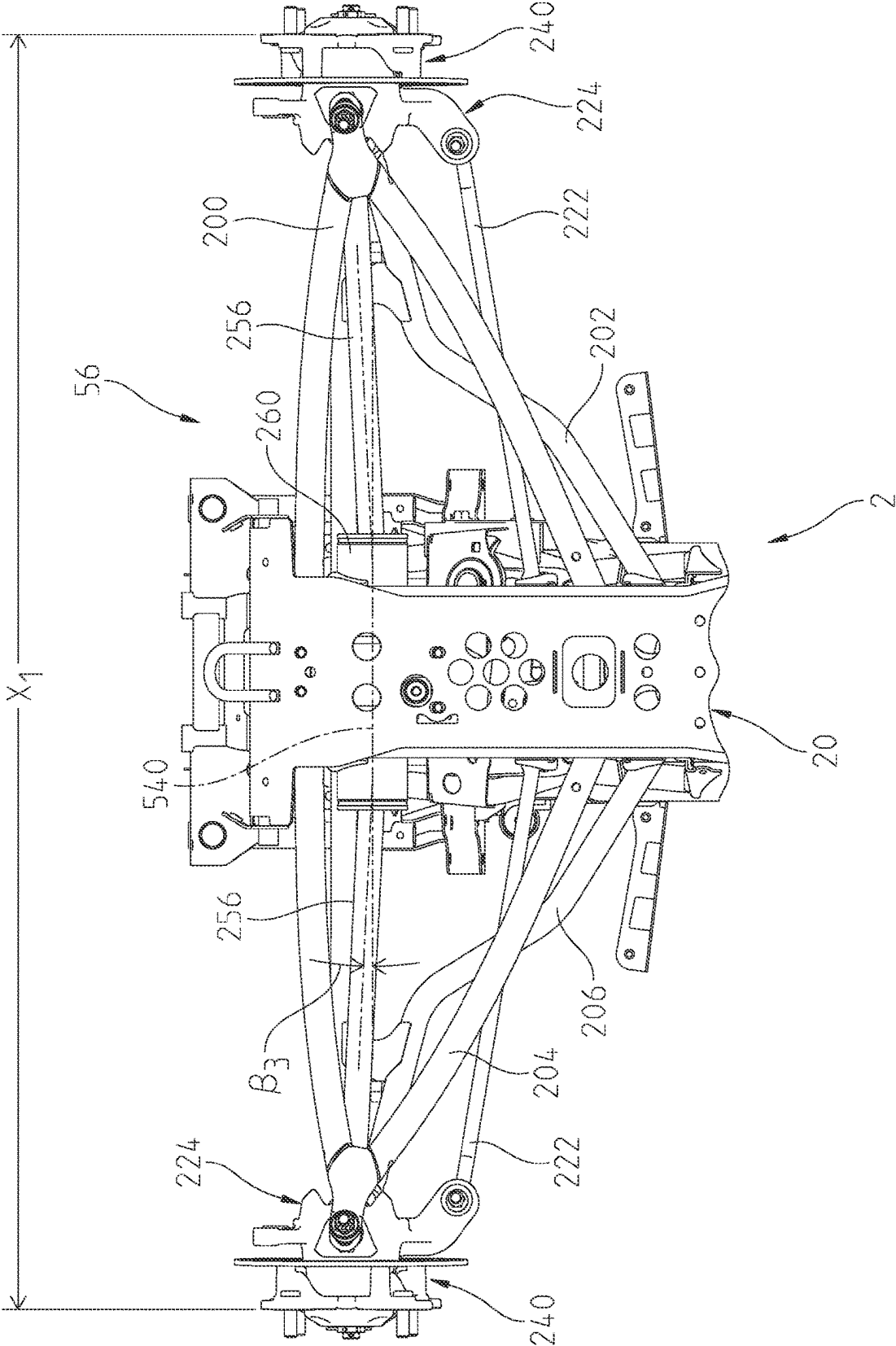


Fig. 21



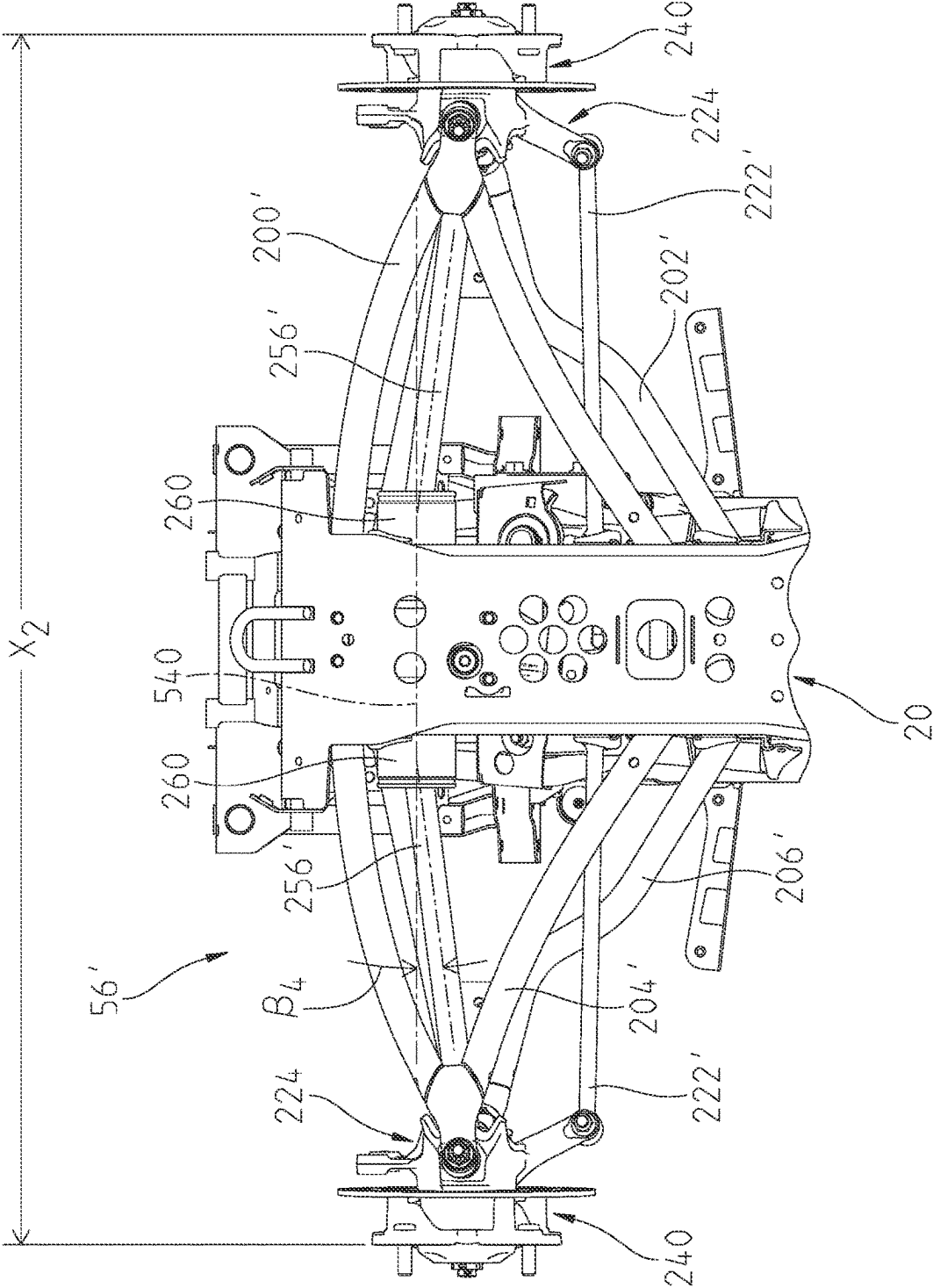


Fig. 23

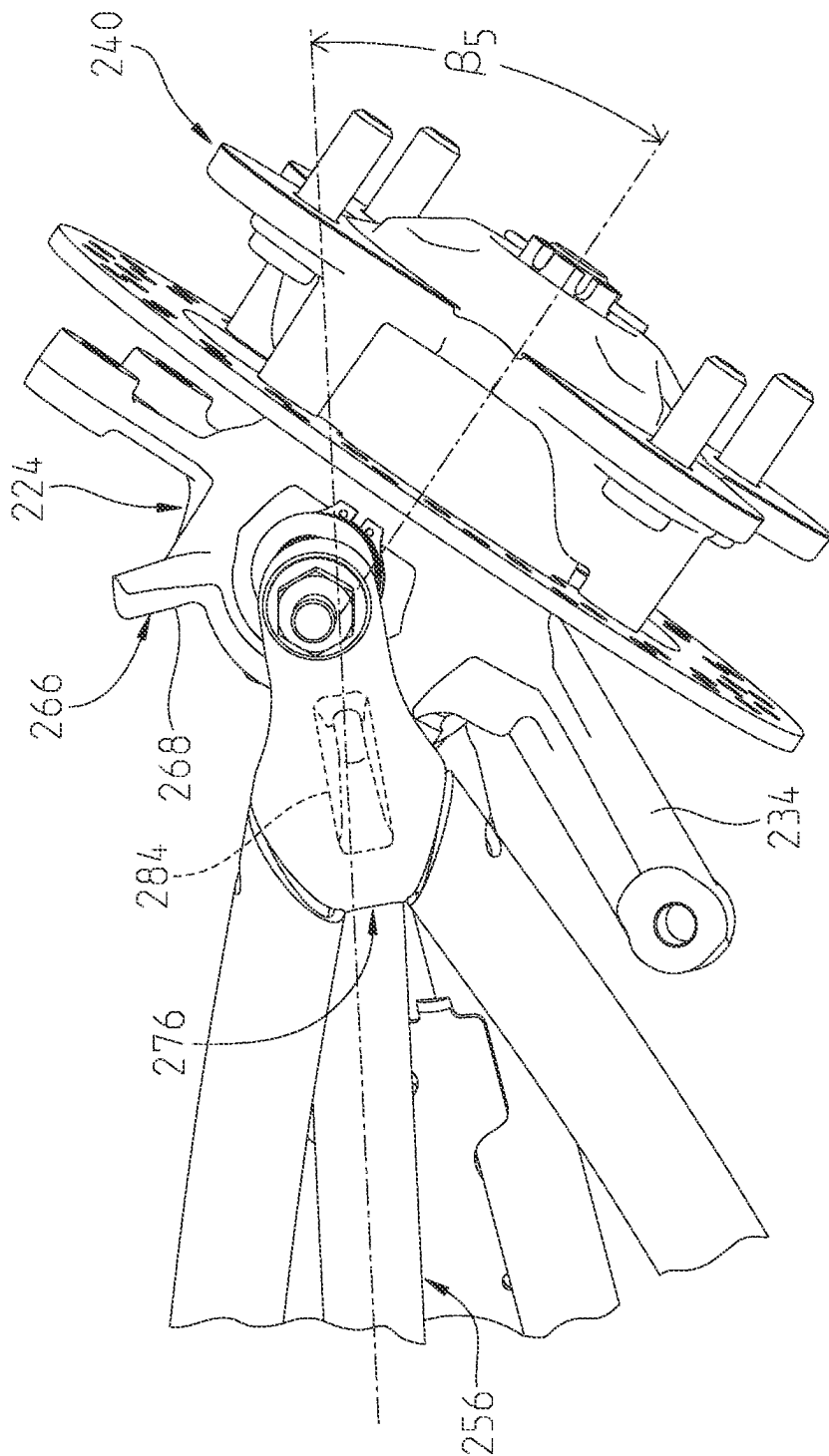


Fig. 24

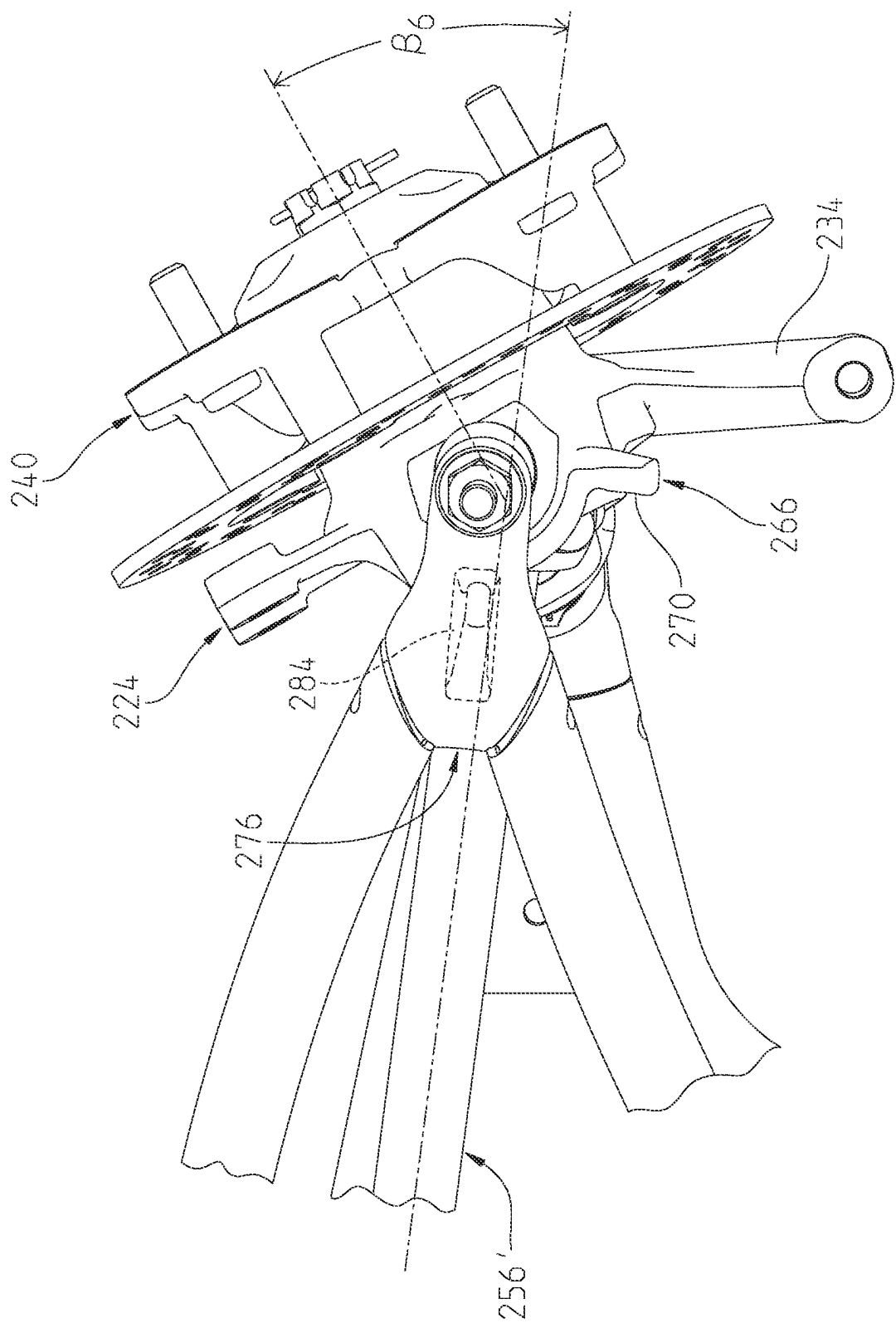


Fig. 25

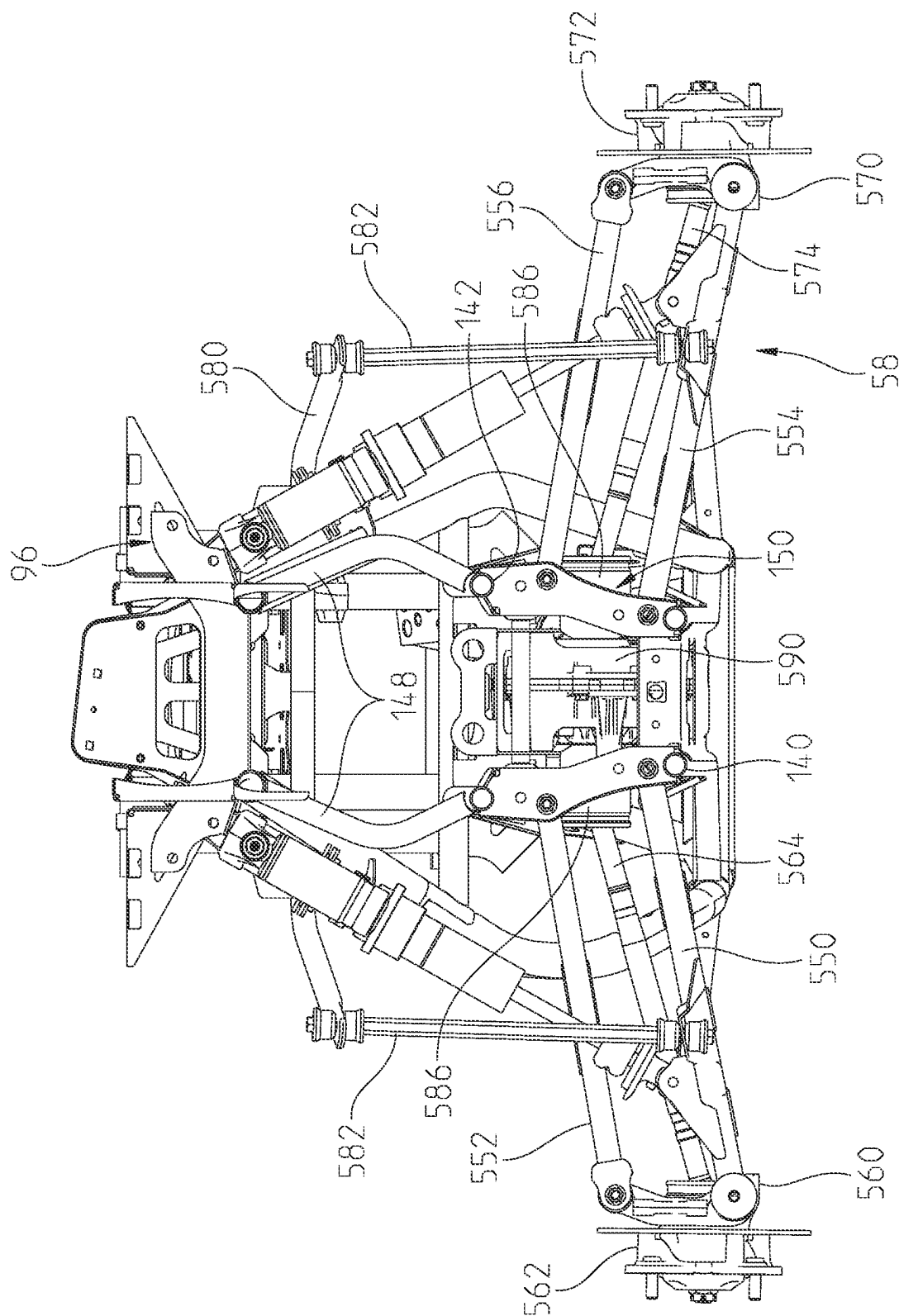


Fig. 26

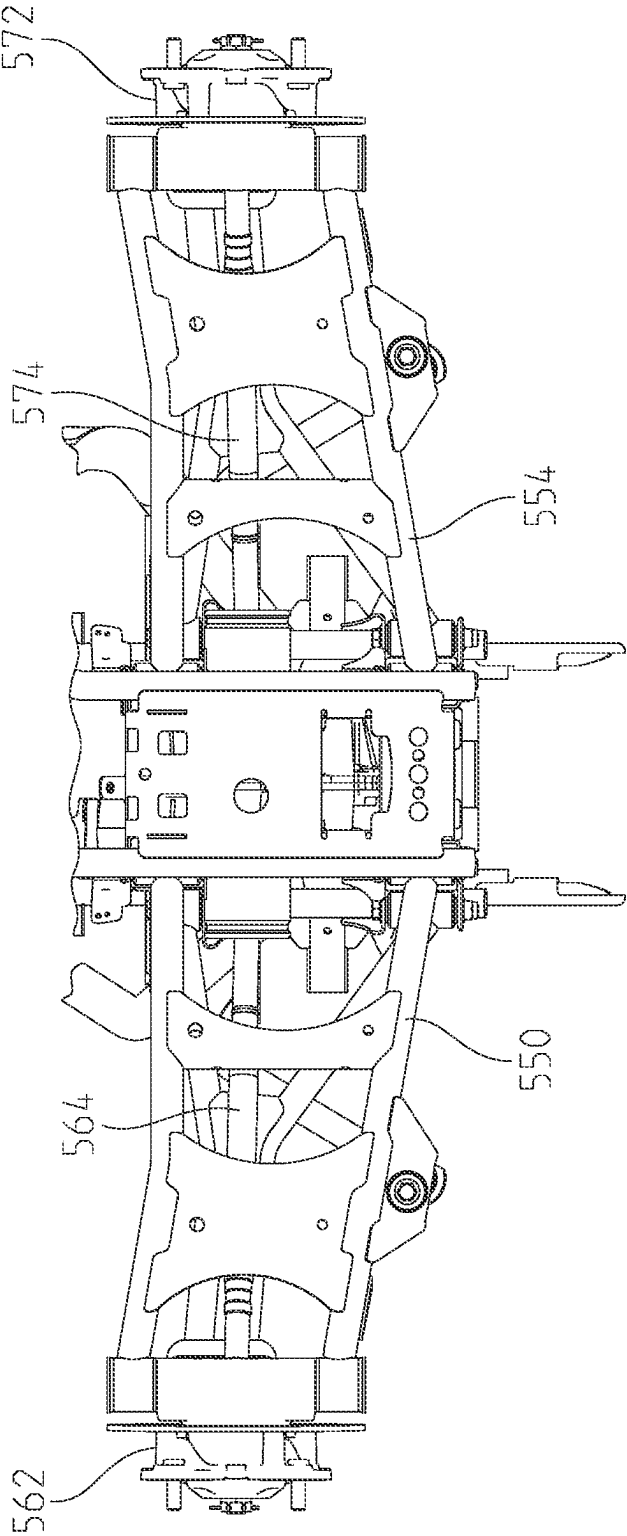


Fig. 27

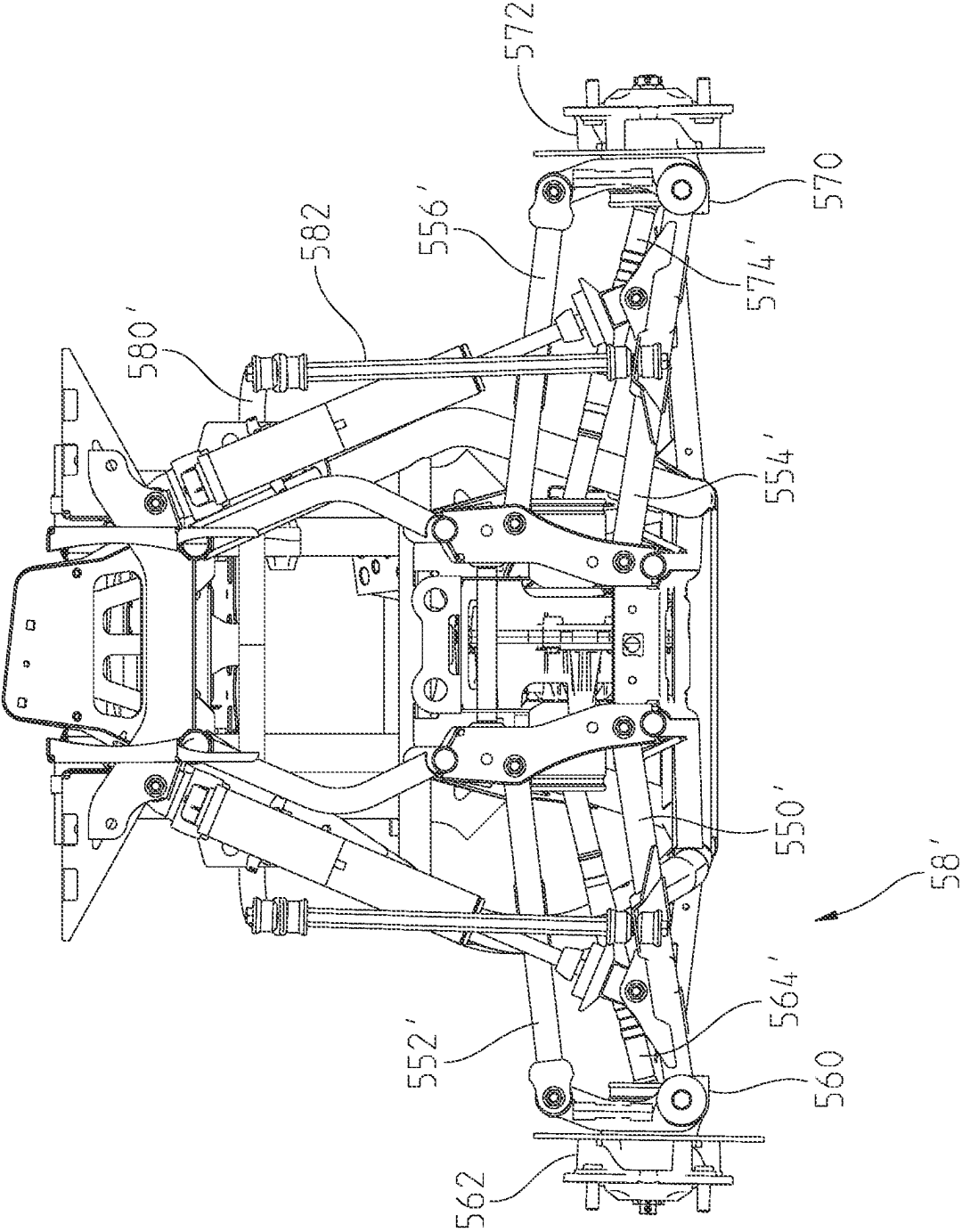


Fig. 28

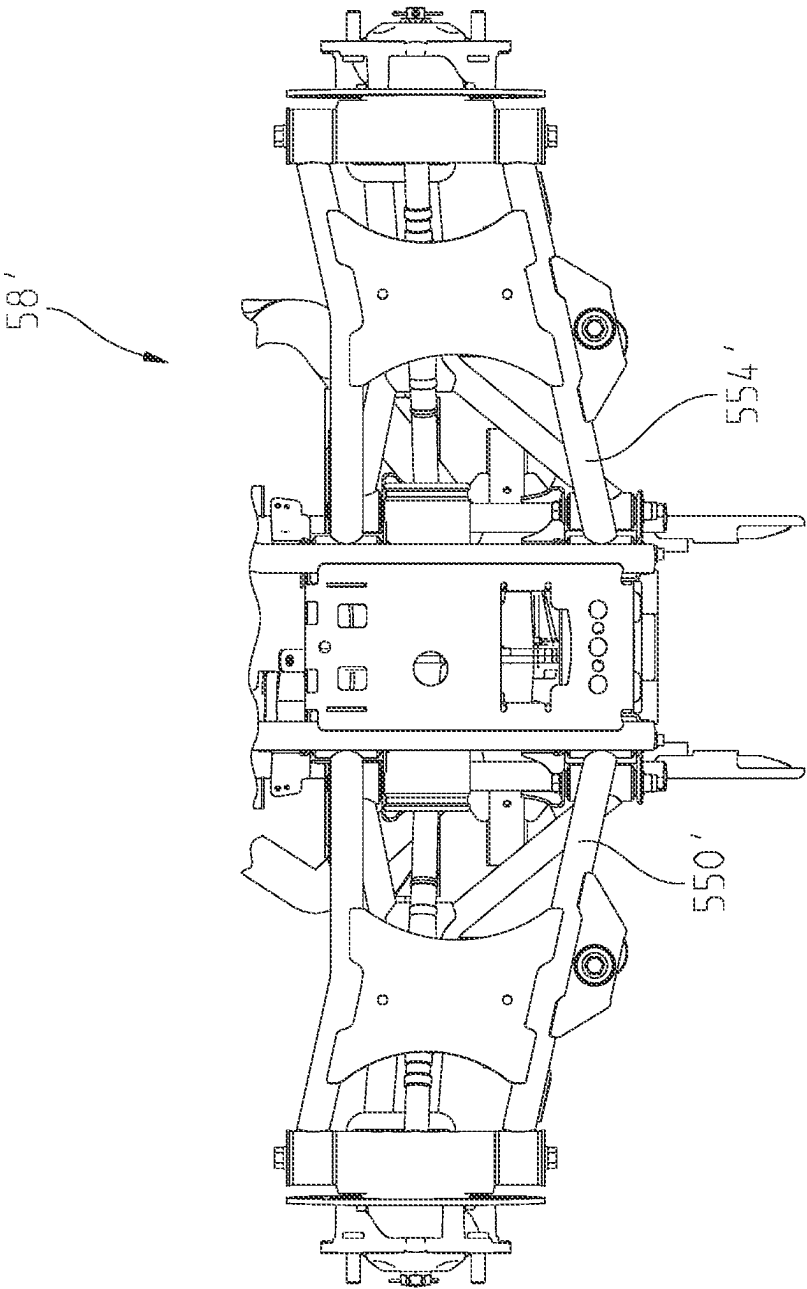


Fig. 29

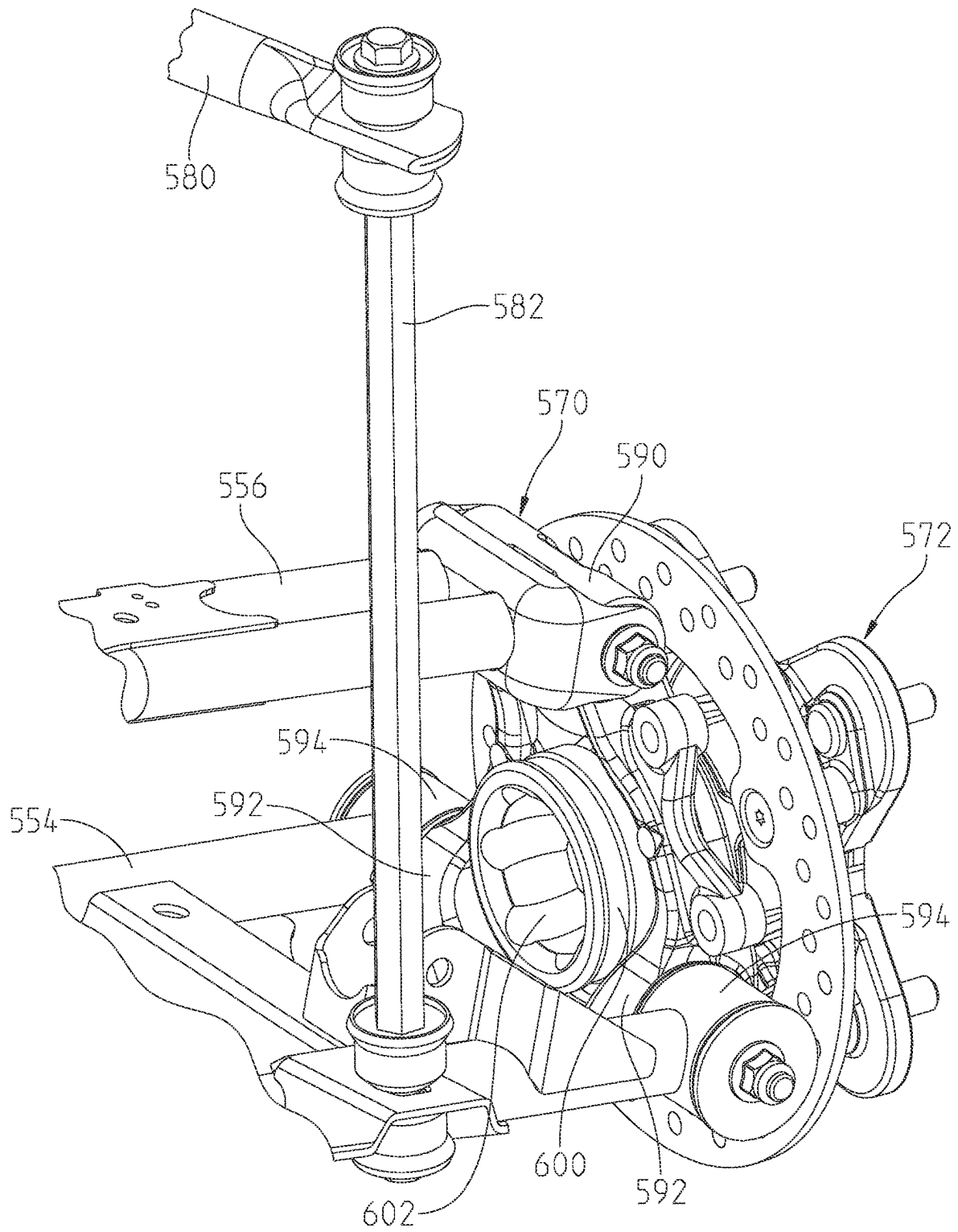


Fig. 30

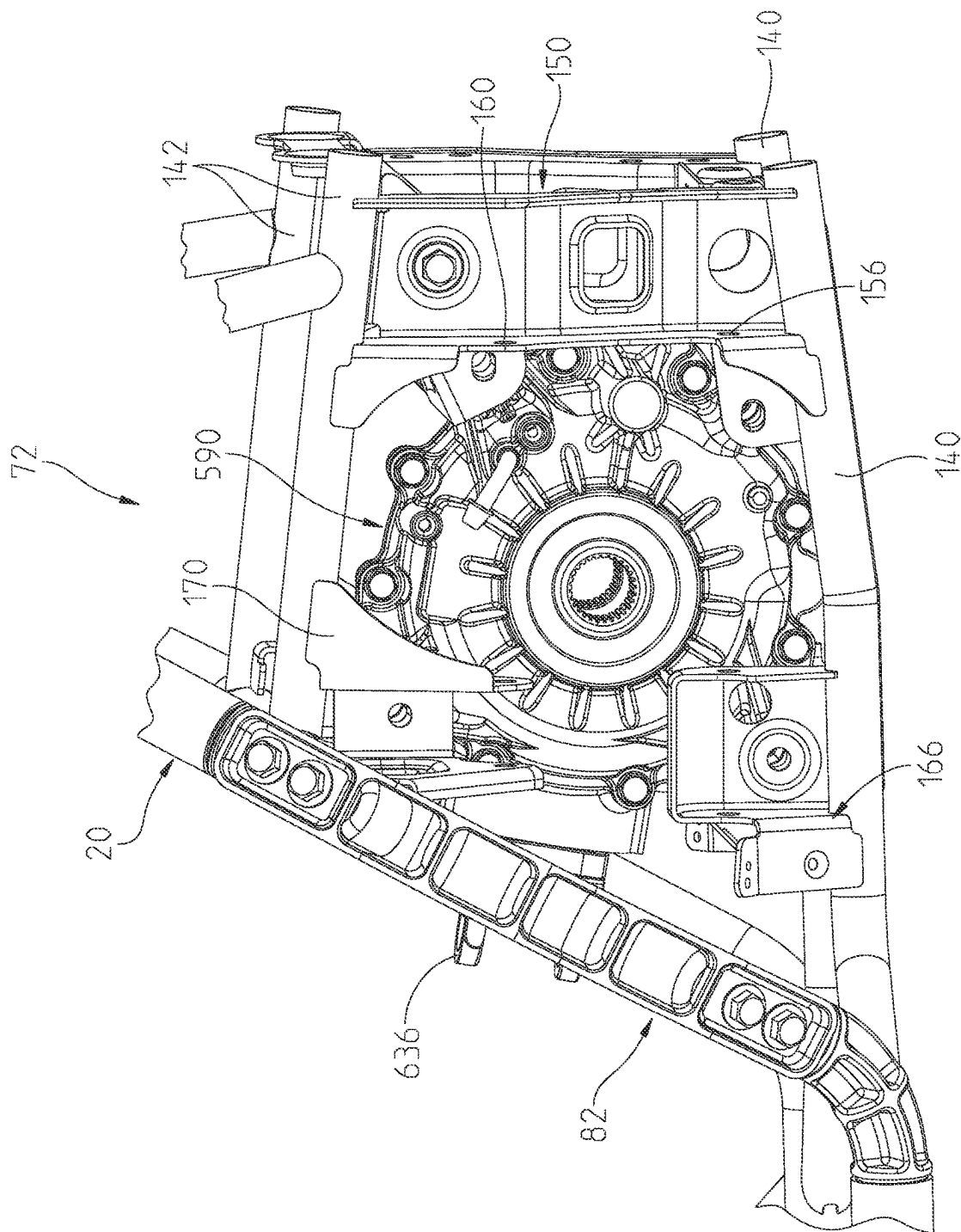
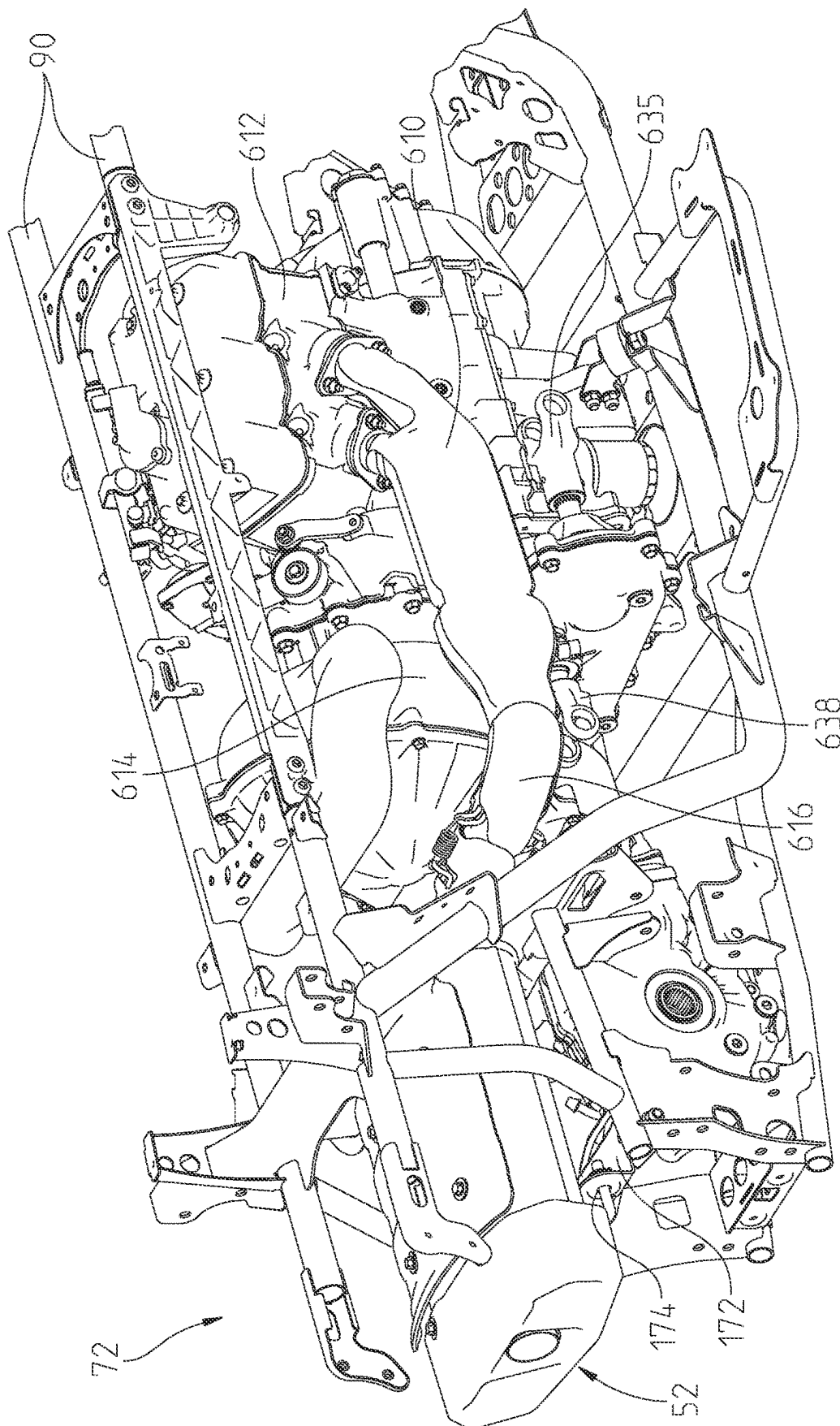


Fig. 31



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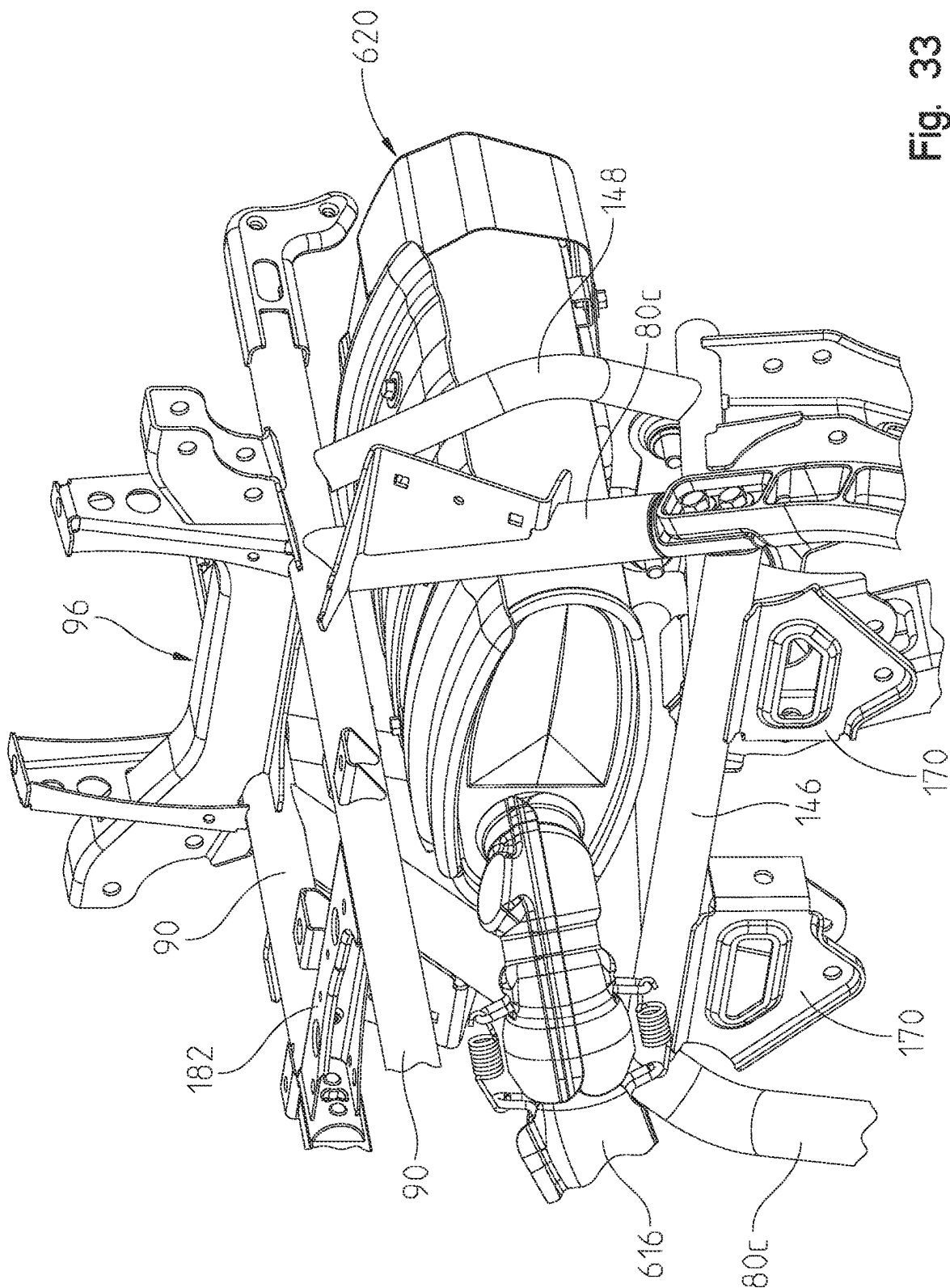


Fig. 33

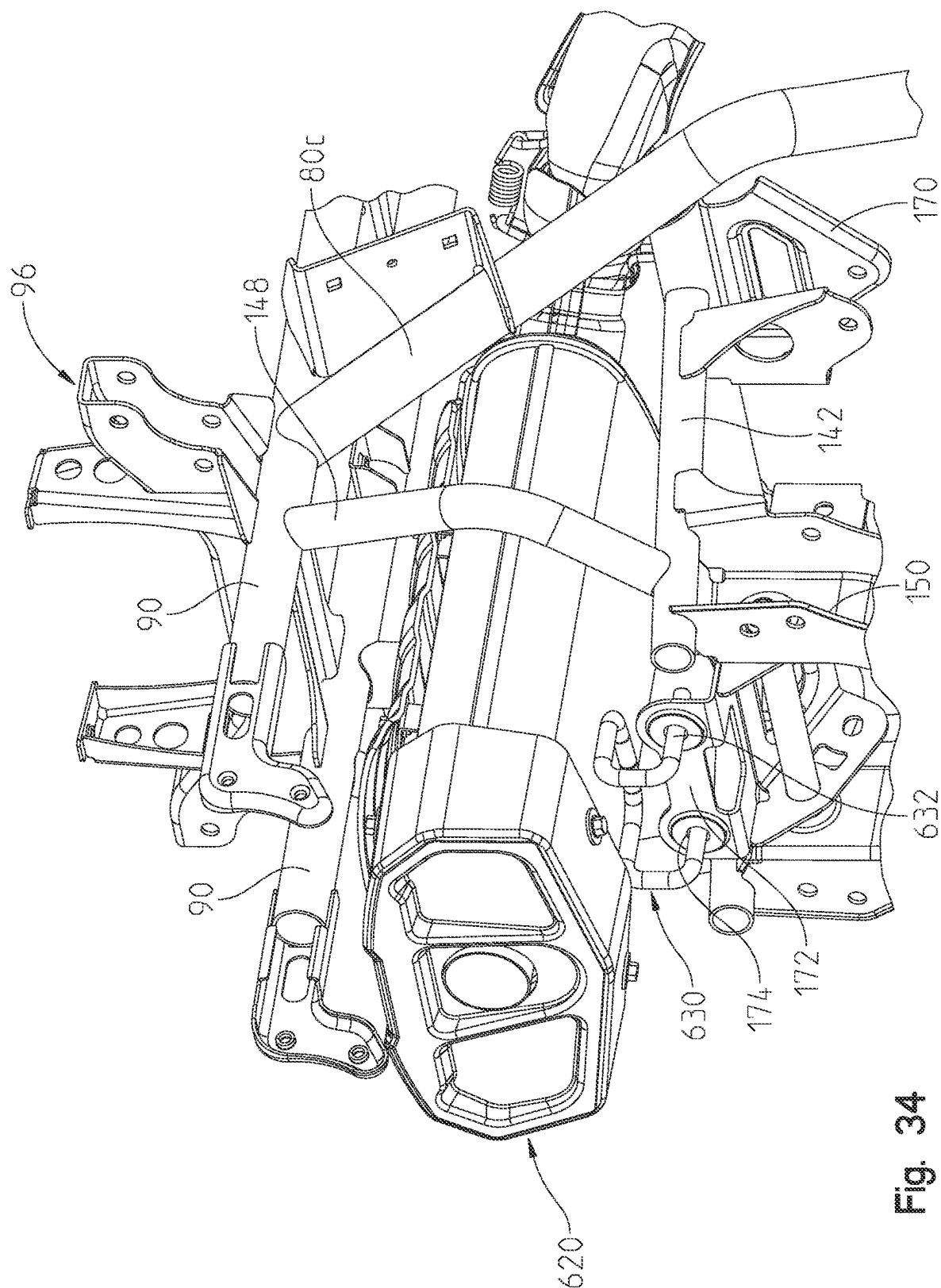


Fig. 34

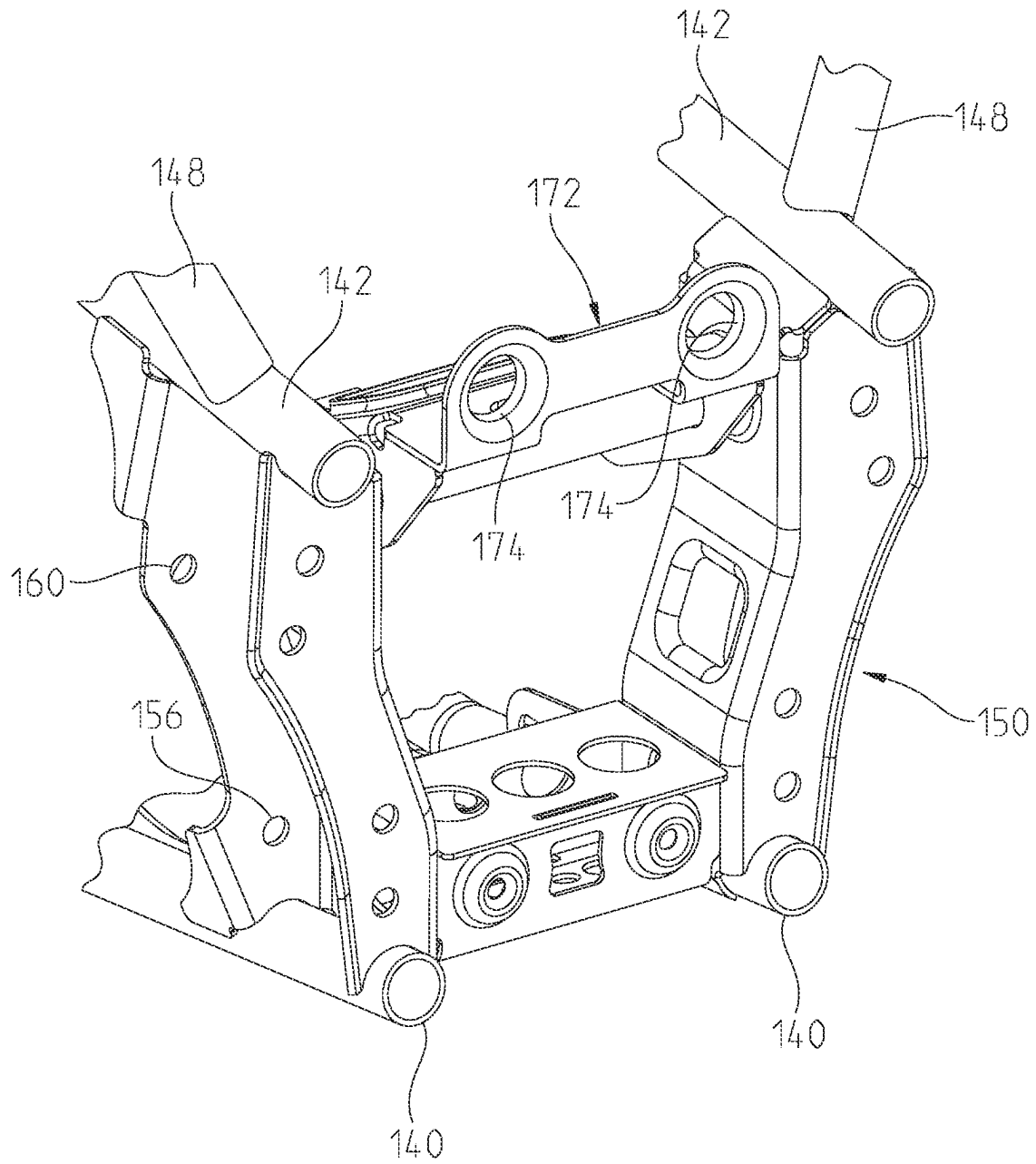


Fig. 35

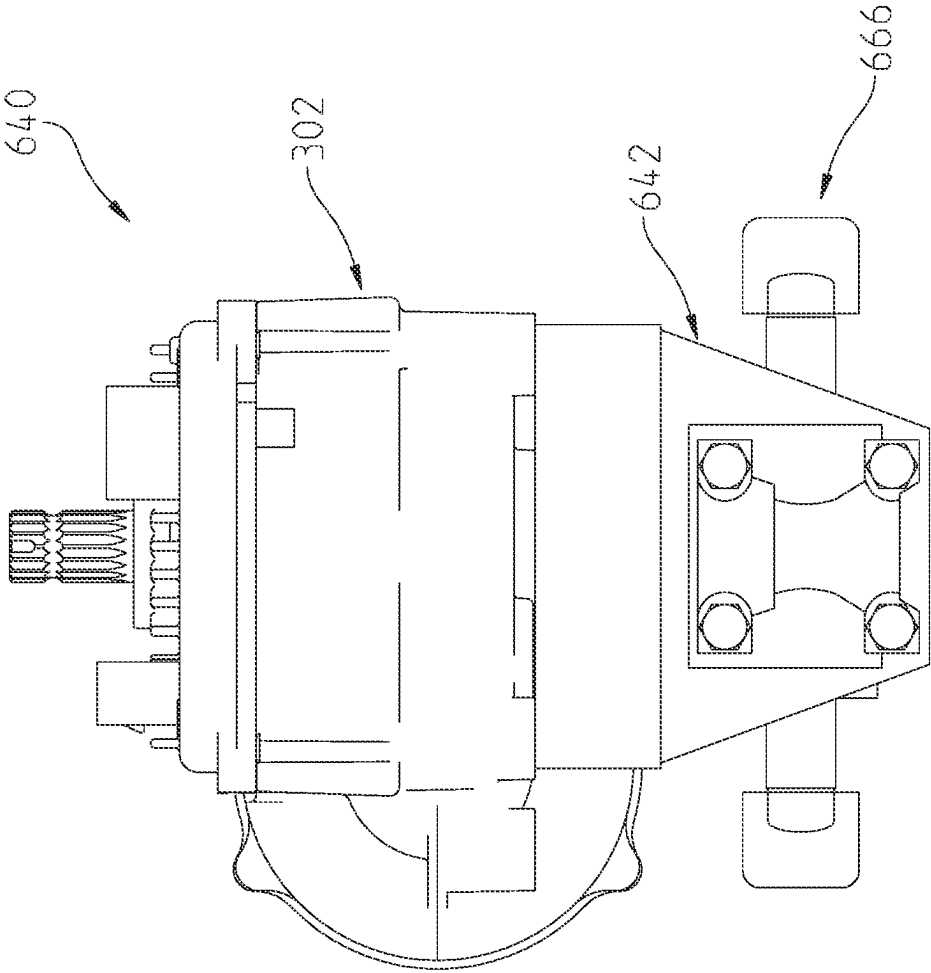


Fig. 36

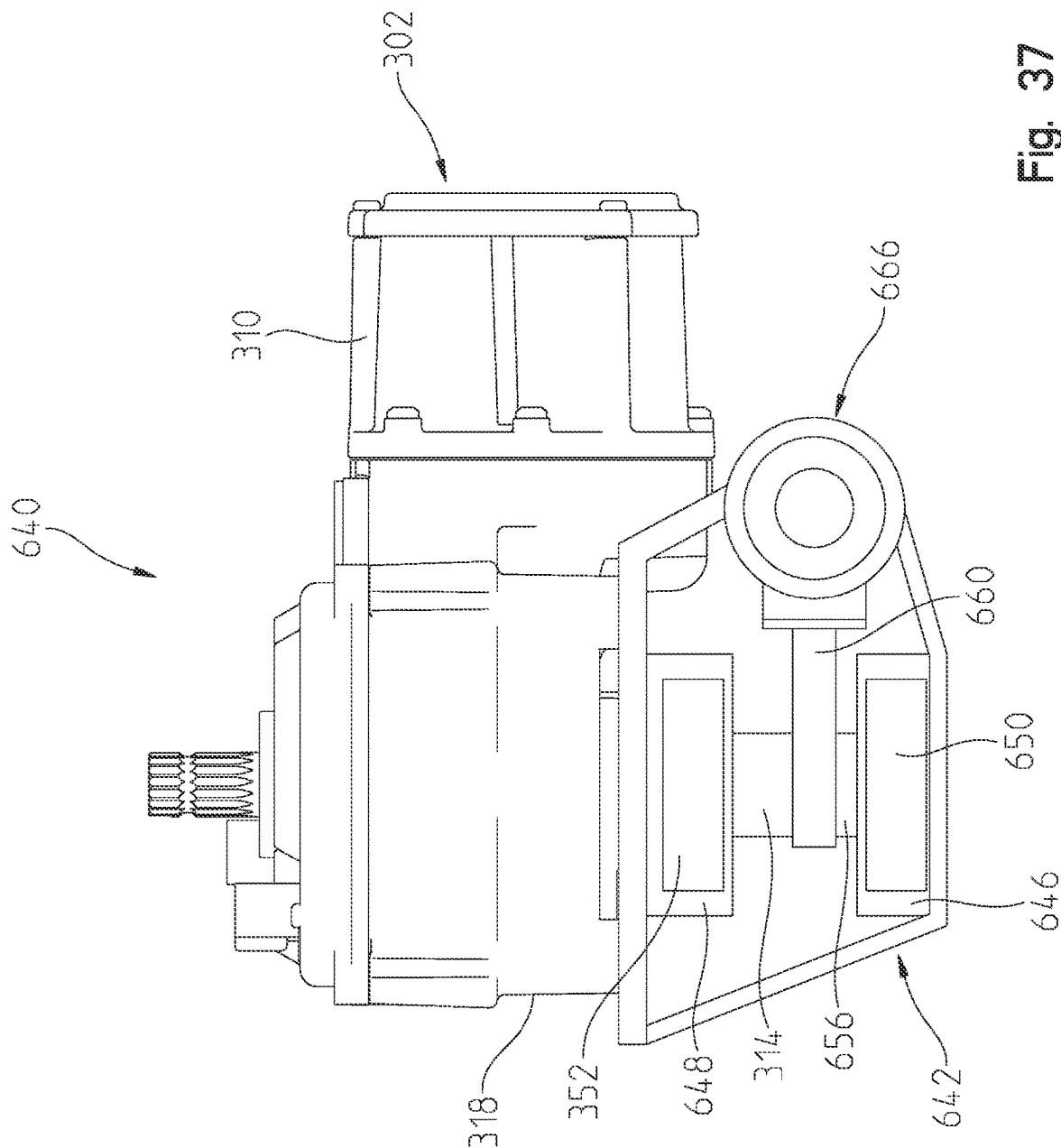


Fig. 37

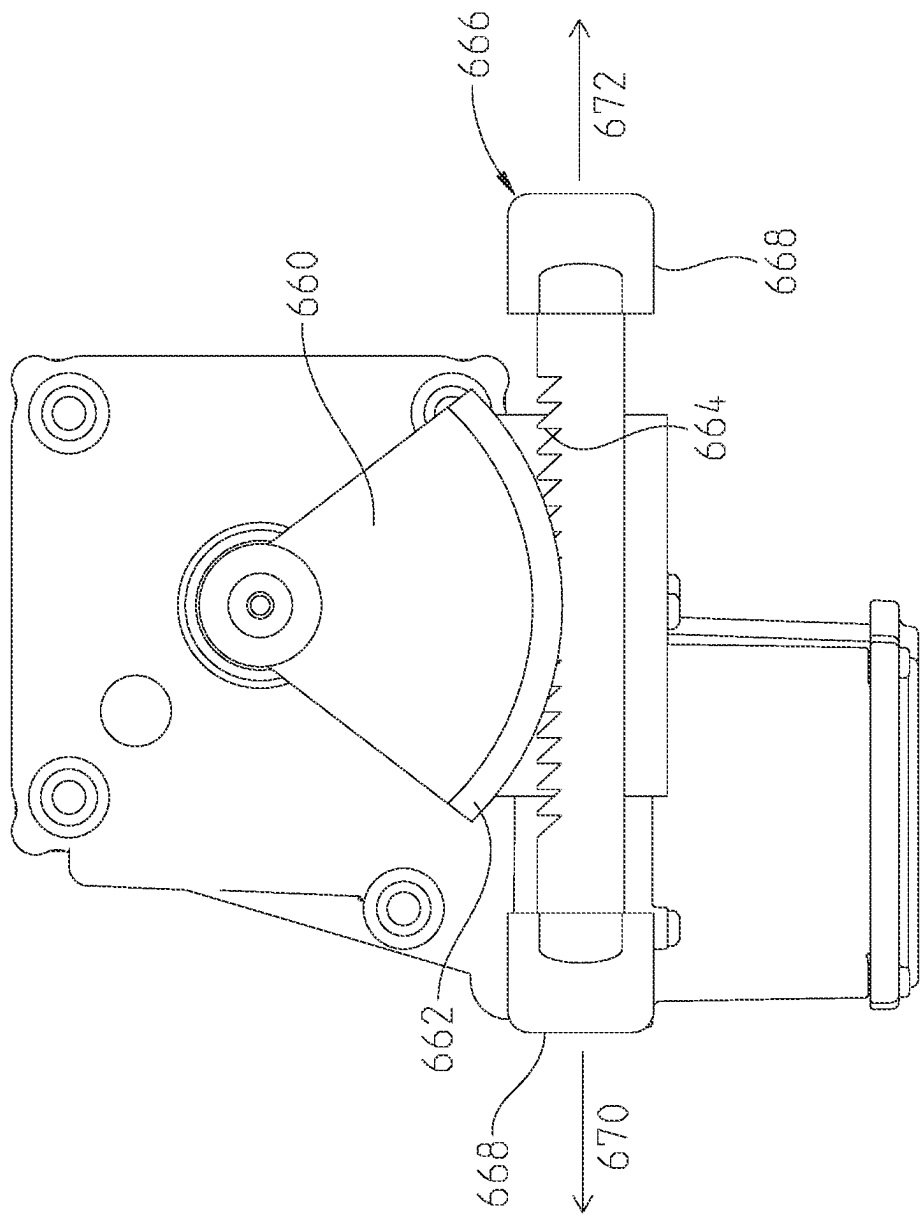


Fig. 38

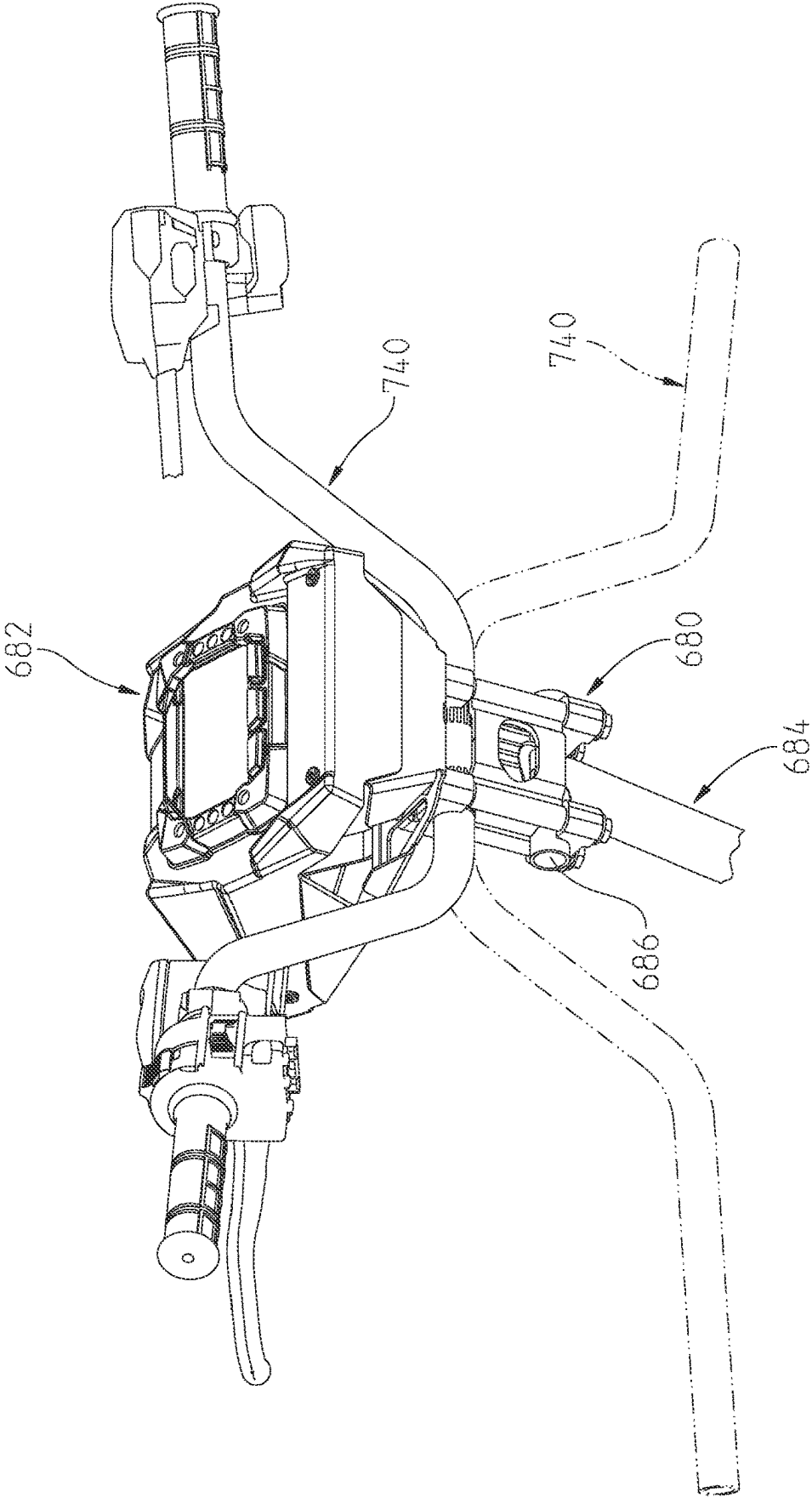


Fig. 39

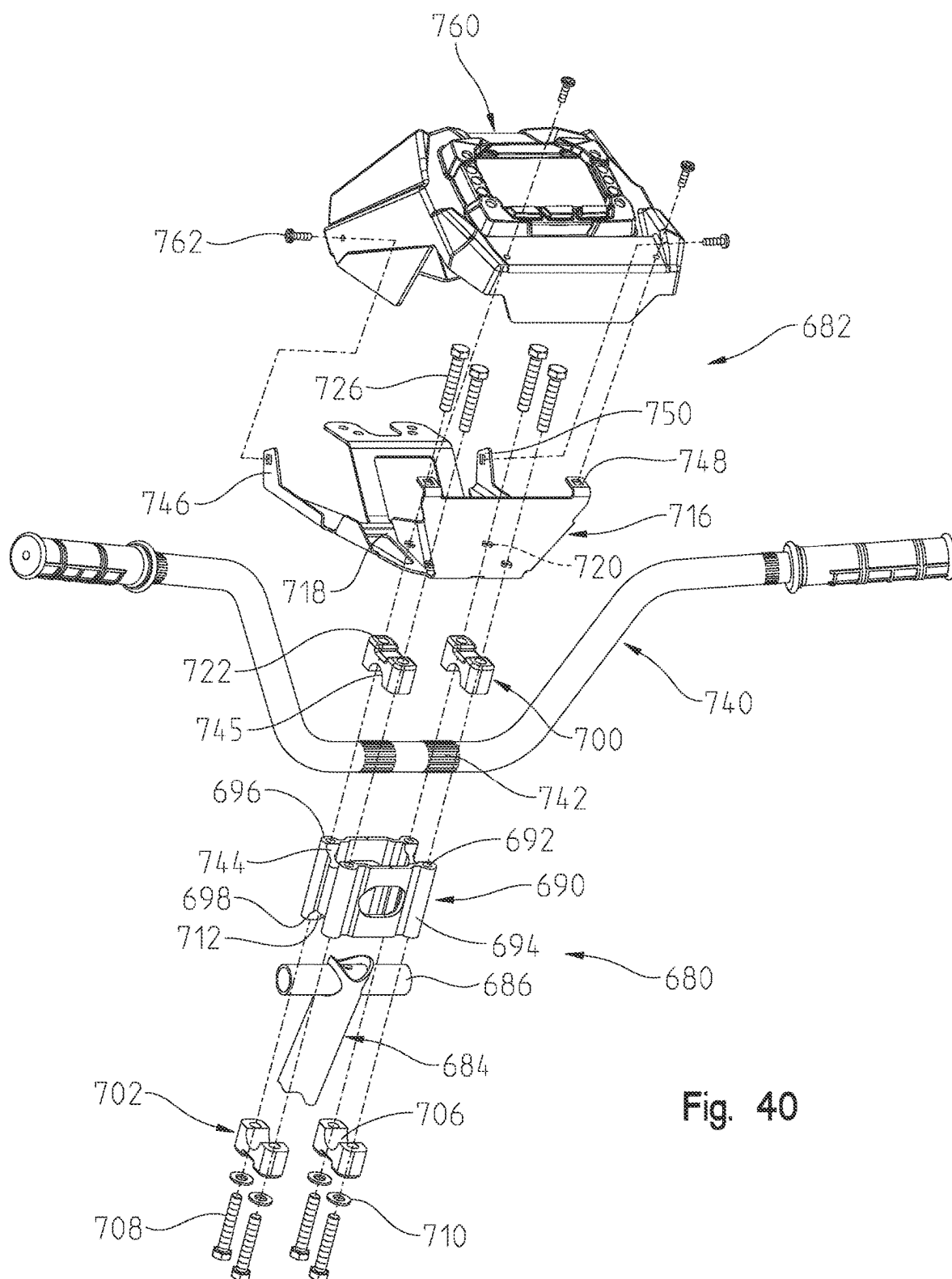


Fig. 40

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ALL-TERRAIN VEHICLE**CROSS-REFERENCE TO RELATED APPLICATIONS**

The present application is a continuation of U.S. patent application Ser. No. 17/062,105, filed Oct. 2, 2020, which is a continuation of U.S. patent application Ser. No. 15/895,495, filed Feb. 13, 2018, the entire disclosures of which are incorporated herein by reference.

BACKGROUND OF THE DISCLOSURE

The present disclosure generally relates to an ALL TERRAIN VEHICLE (ATV) and, more particularly, to an ATV having improved ergonomics and performance.

Generally, ATVs are used to carry one or two passengers and a small amount of cargo over a variety of terrains. Due to increasing recreational interest in ATVs, specialty ATVs, such as those used for trail riding, racing, and cargo hauling have entered the market place. Most ATVs include an engine including between one and three cylinders. Most ATVs include a straddle or saddle type seat positioned above the engine.

In one example of the invention, a method of making multiple widths of all-wheel drive vehicles is described which includes the steps of providing a frame having a front portion and a rear portion; providing a power train and coupling the power train to the frame, the power train comprising a primary motive unit, a transmission, a front final drive, a rear final drive and couplings between the transmission and the front and rear final drives; providing a plurality of front and rear suspension components of different lengths; providing a plurality of front and rear half shafts of different lengths; selecting sets of front and rear suspension components and front and rear half shafts to define a measured distance from a longitudinal centerline of the vehicle; and providing front and rear wheels, where a distance measured from outside the front and rear wheels is different for the different sets.

In another illustrative embodiment, an all-terrain vehicle (ATV) comprises a frame; ground engaging members supporting the frame, including front and rear ground engaging members; front right and left suspensions; front wheel steering mechanisms including right and left knuckles pivotally coupled to the respective front right and left suspension; a power steering unit coupled to the frame, the power steering unit including an output shaft which is laterally offset from a vertical centerline of the vehicle; right and left steering arms coupled to respective right and left knuckles, and a linkage coupling the output shaft to the steering arms, where the right and left steering arms are substantially the same length.

In another illustrative embodiment, an all-terrain vehicle (ATV) comprises a frame; ground engaging members supporting the frame, including front and rear ground engaging members; front right and left suspensions; front wheel steering mechanisms including right and left knuckles pivotally coupled to the respective front right and left suspension; a first stop member positioned on each of the front right and left suspension; and a second stop member positioned on each of the right and left knuckles; wherein the first and second stop members cooperate to provide extreme rotational positions.

In another embodiment includes an all-terrain vehicle (ATV) comprising a frame; ground engaging members supporting the frame, including front and rear ground engaging

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members; front right and left suspensions; a power steering unit coupled to the frame; a steering post providing torsional input to the power steering unit; a steering system support bracket coupled to the frame for supporting the power steering unit; and a steering stop coupled between the steering shaft and the steering system support bracket, to provide extreme rotational positions of the steering shaft.

Finally, in another embodiment, an all-terrain vehicle (ATV) comprises a frame; ground engaging members supporting the frame, including front and rear ground engaging members; front right and left suspensions; a power steering unit coupled to the frame, the power steering unit comprising a power steering motor having an output shaft and a rack and pinion subassembly coupled to the power steering unit, the rack and pinion subassembly having a housing and a pinion gear driving coupled by the output shaft, and a rack drivably coupled to the pinion gear; a steering post providing torsional input to the power steering unit; and steering arms coupled to the rack; wherein torsional input to the power steering unit causes rotation of the output shaft and pinion, and linear movement of the rack and steering arms.

The invention will now be described by way of reference to the drawing figures, where:

FIG. 1 is a front left perspective view of the ATV of the present disclosure;

FIG. 2 is a right rear perspective view of the ATV of FIG. 1;

FIG. 3 is a left side view of the ATV;

FIG. 4 is a right side view of the ATV;

FIG. 5 is a top view of the ATV;

FIG. 6 is a front view of the ATV;

FIG. 7 is a rear view of the ATV;

FIG. 8 is a front left perspective view of the frame for the ATV of FIGS. 1-7;

FIG. 9 is a right rear perspective view of the frame of FIG. 8;

FIG. 10 is a front left perspective view of a 55" wide version of the ATV showing the front suspension and steering system;

FIG. 11 is a front view of the ATV front suspension and steering system of FIG. 10;

FIG. 12 is an enlarged rear view of the right-hand suspension system of FIG. 11;

FIG. 13 is a front left perspective view of the wheel hub shown in FIG. 12;

FIG. 14 is an exploded view of the wheel hub coupling to the lower A-arm of FIG. 13;

FIG. 15 is a right rear perspective view of the ATV shown in FIG. 10;

FIG. 16 is an underside perspective view of the suspension and steering system;

FIG. 17 is an underside perspective view showing the electric power steering unit mounting and the steering arm linkage;

FIG. 18 is an exploded view of the electric power steering unit mounting and steering linkage of FIG. 17;

FIG. 19 shows a rear right perspective view of the steering linkage of FIG. 18;

FIG. 20 is a rear left perspective view of the linkage of FIG. 19;

FIG. 21 is a lower perspective view showing the driven pitman arm together with the steering stop;

FIG. 22 is an underside view showing the suspension of a 55" wide ATV;

FIG. 23 shows a view similar to that of FIG. 22 showing the suspension of a 48" wide ATV;

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FIG. 24 shows a bottom view of the left side suspension of a 55" wide ATV;

FIG. 25 shows a bottom view of the left side suspension of a 48" wide ATV;

FIG. 26 shows a rear view of the rear suspension of a 55" wide ATV;

FIG. 27 shows a bottom view of the suspension of FIG. 24;

FIG. 28 shows a rear view of the 48" rear suspension;

FIG. 29 shows a bottom view of the suspension of FIG. 26;

FIG. 30 shows a right rear perspective view of the right rear wheel hub;

FIG. 31 shows a left rear perspective view of the mounting of the rear final drive and the frame of the ATV;

FIG. 32 shows a right rear perspective view showing the engine mounting together with the exhaust routing for the ATV frame;

FIG. 33 shows a front left perspective view of the ATV muffler of FIG. 32;

FIG. 34 shows a right rear view of the muffler of FIG. 31;

FIG. 35 shows a rear left perspective view of the hitch mount on the ATV;

FIG. 36 is a front view of an alternate rack and pinion steering system;

FIG. 37 is a left hand side view of the rack and pinion steering system of FIG. 36;

FIG. 38 is a bottom view of the rack and pinion steering system of FIG. 36;

FIG. 39 is a left rear view of a riser block assembly usable with the handle bars; and

FIG. 40 is an exploded view of the assembly of FIG. 39.

Corresponding reference characters indicate corresponding parts throughout the several views. Unless stated otherwise the drawings are proportional.

DETAILED DESCRIPTION OF THE DRAWINGS

With reference first to FIGS. 1-7, the ATV of the present disclosure will be described in greater detail. As shown, ATV is shown generally at 2 and comprises front wheels 4 and rear wheels 6. Front wheels 4 include rims 8 and tires 10, whereas rear wheels 6 include rims 12 and tires 14. Wheels 4 and 6 support a frame 20 which in turn supports a front body portion 22, a rear body portion 24 including a rack 26, a seat 28 and a mid-body portion 30 including a foot well 32. ATV 2 includes a steering system 36 including handlebars 38 for steering ATV 2, as described herein. A pod 40 is coupled to handlebars 38 and includes a headlight 42 (FIG. 1) and a control display 44 (FIG. 2). ATV 2 further comprises a powertrain 50 (FIG. 3) including an exhaust system 52 (FIG. 2). Finally, ATV 2 includes a front suspension system 56 (FIG. 6) and a rear suspension 58 (FIG. 7).

As will be disclosed herein, ATV 2 can be provided in two widths, that is a 48" width and a 55" width. In accordance with the present disclosure, the frame 20, body components 22, 24 and 30, and powertrain 50 are identical for the two vehicle widths. Rather, the front and rear suspensions 56, 58 interchange to provide the wider vehicle, as disclosed herein.

With reference now to FIGS. 8 and 9, frame 20 will be described in greater detail. As shown, frame 20 includes a front frame portion 70, a rear frame portion 72, a foot support portion 74 and a seat support portion 76. Frame 20 is generally defined by longitudinally extending tubes 80 having horizontal sections 80a, which turn upwardly and forwardly at portions 80b and turn upwardly and rearwardly

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at portions 80c. A removable forging 82 is provided on the driver's side of the vehicle, such that removal provides an access opening for installing or servicing components of the ATV. Foot support portion 74 is defined by a U-shaped tube 84 coupled to the two portions 80a and including mounting brackets 86. Seat support portion 76 is defined by longitudinally extending tubes 90 which couple at the top ends of frame tube portions 80b and 80c. Front ends of tubes 90 terminate in a cross brace 94, while tubes 90 support a rear cross brace 96.

Front frame portion 70 includes front tubes 96 extending upwardly from a cross brace 98 to couple to a front side of cross brace 94. Upright tubes 100 extend upwardly from cross brace 98 and couple with tubes 96. Tubes 102 extend longitudinally between tube portions 80b and tubes 100, and support a steering system support bracket 110, as described further herein. Bracket 110 includes a lower bracket portion 112 fixed to tubes 102, for example, by welding. Frame 20 further includes lower tubes 114 which couple between tubes 80 and a front brace 116, where brace 116 includes an upright portion 118 which couples to cross brace 98. Four attachment brackets are provided for mounting of the front suspension 56, namely, a front lower bracket 126, a lower rear bracket 128, an upper front bracket 130 and an upper rear bracket 132. These same brackets 126, 128, 130 and 132 are replicated on the right hand side in a mirror image configuration.

Rear frame portion 72 includes lower tubes 140 coupled to tubes 80a which extend rearwardly, while longitudinal tubes 142 extend rearwardly from a cross tube 146. Vertical tubes 148 extend upwardly between tubes 142 and 90. Braces 150 extend upwardly between tubes 140 and 142 and provide a first attachment point 156 (FIG. 9) for a rear lower coupling for the rear suspension 58 and an upper coupling point 160 for the rear suspension 58. A front lower bracket 166 is coupled to each of the tubes 140 for a front lower coupling point of rear suspension 58, while an upper bracket 170 provides coupling for a front upper coupling point for rear suspension 58. A muffler support 172 is positioned at a rear of the frame 20 and straddles frame members 142, and includes apertures 174.

Finally and with respect to seat support portion 76, a forging 180 is provided, interrupting a length of the passenger side tube 90 which couples to cross braces 182 and which is removable for the assembly and/or servicing of the vehicle components, such as powertrain components.

With reference now to FIGS. 10-16, the front suspension 56 of the 55" vehicle will be described. With reference first to FIG. 10, a front end of the vehicle is shown without the wheels and the front body 22 which exposes the front suspension 56 and frame 20. A front drive 190 is also shown to drive the front wheels 4 from the powertrain. As is known, at the front end of the vehicle, the steering system 36, the front suspension 56 and the front drive 190 all cooperate together.

As shown in FIG. 10, front suspension 56 includes a lower left A-arm 200, upper left A-arm 202, lower right A-arm 204 and upper right A-arm 206. Shock absorbers 208 and 210 couple between upper A-arms 202, 206 and upper brace 94, respectively. Inner ends of lower left A-arm 200 couple to coupling points 126, 128 (FIG. 8), while inner ends of upper left A-arm 202 couple to coupling points 130, 132 (FIG. 8). Right A-arms 204, 206 are likewise coupled to like coupling points of the frame 20 on the right-hand side. As shown best in FIG. 12, upper A-arms 202 and 206 have an upwardly directed bend at 220 which is profiled to provide clearance for steering arm 222. As will be discussed further herein,

steering arm 222 is coupled at an inner end to a motor of the steering system 36, and is coupled at its outer end to knuckle 224. Knuckle 224 is coupled to A-arms 204, 206 by way of pins 230, 232. Knuckle 224 includes an extension 234 which couples to threaded posts 236 of steering arm 222. Wheel hubs 240 are provided having studs 242 for mounting wheels 4.

As shown best in FIG. 13, hub 240 includes a constant velocity joint 250 having an inner profiled geometry 252 which is suitable for driving by way of a half shaft (or constant velocity (CV) shaft) 256 (FIG. 10), as is known in the art. Inner ends of half shafts 256 likewise include drive portions to be driven by constant velocity couplings 260 (FIG. 10) from the drive unit 190. As shown best in FIGS. 13 and 14, knuckle 224 includes an integrated stop member 266 in the form of an arcuate slot having extreme contact surfaces 268 and 270 (FIG. 14). Meanwhile, lower right A-arm 204 includes a coupling 276 having an opening at 278 which is received over pin 230 to receive a castellated nut 280 (FIG. 13) to couple lower A-arm 204 to knuckle 224. As shown, coupling 276 includes an integrated stop member 284 defining an upstanding post having a shark fin geometry, whereby hub 224 can rotate about an axis defined by pins 230, 232 to the extreme positions where surface 268 or 270 contacts stop member 284.

The forging also makes contact at a 45° angle instead which increased the strength significantly. This design is lighter and gives greater control to the stop. The shark fin 284 is wider at the base 284a, as compared to the top 284b, (see FIG. 14) which allows a progressive steering stop and limits the steering angle as the wheel travels downward to protect the half shafts 256 (FIG. 10) at suspension full extension.

With reference now to FIGS. 15-21, steering system 36 will be described in greater detail. With reference first to FIG. 15, steering system 36 includes a steering post 300 extending downwardly from handlebars 38 (FIG. 1) to a power steering motor 302. As shown best in FIG. 17, steering system support bracket 110 further comprises an upper bracket portion 304. Steering motor 302 is coupled to bracket portion 304 (FIG. 16), which in turn couples to lower bracket portion bracket 112 (FIG. 15). With reference now to FIGS. 17 and 18, the mounting of steering motor 302 will be described in greater detail. As shown best in FIG. 18, power steering motor 302 includes a motor portion 310 and an output portion 312 having a splined drive shaft 314 extending through a bearing cap 316. Four threaded bosses 318 are provided for coupling to bracket 304. As shown, bracket 304 includes an upper plate 320 having a bracket tab 322 extending from one end thereof having apertures 324 and 326. At the opposite end, a bracket sidewall is provided at 330 having apertures at 332 (FIG. 17). Plate portion 320 includes an enlarged aperture at 336 (FIG. 19) which fits over bearing cap 316, as best shown in FIG. 17. Plate portion 320 further includes apertures 340 (FIG. 19) which align with the threaded bosses 318 to receive fasteners 344 therethrough coupling bracket 304 to a lower side of steering motor 302, as shown best in FIG. 18.

As should be appreciated, when bracket 304 is coupled to the bottom of steering motor 302, splined shaft 314 protrudes through aperture 336. A drive pitman arm 350 is provided having a coupling 352 having an inner spline 356 (FIG. 19). Pitman arm 350 includes an arm portion 360 having an aperture 362 adjacent a free end of the arm 360 and an aperture 364 intermediate aperture 362 and shaft portion 370. Shaft portion 370 is received in a bearing assembly 376 having a plate portion 378 (FIG. 21) including

a rotatable stop portion 380 having stop surfaces 380a and 380b with a bearing 384 extending therethrough. Bearing assembly 376 can be coupled to lower bracket 110 as described herein. Lower bracket 112 includes a plate portion 400 (FIG. 18) having an aperture at 402 to receive bearing portion 384. Plate portion 400 includes square apertures 404 which align with square apertures 406 on bearing assembly 376 which, as can be appreciated, receive carriage bolts therethrough to couple the bearing assembly 376 to the plate portion 400.

As shown best in FIG. 21, drive pitman arm 350 includes a stop member 410 which rotates between the extreme portions of surfaces 380a and 380b under the influence of splined shaft 314. Finally, drive pitman arm 350 includes a groove 416 (FIG. 21) to receive a snap ring 418 (FIG. 18), therein which retains the shaft portion 370 in the bearing 384, and the pitman arm 350 to the splined shaft 314. The steering stop 380, 410 on the post protect the CV shafts 256 primarily, the post stop protects the wheels 4 from contacting the bodywork during neutral suspension position. The plate 380 is interchangeable and allows adjustment of the stop position which is important at the neutral position when larger tires are desired.

As shown best in FIG. 18, a driven pitman arm is shown at 420 which is similarly constructed as pitman arm 350, however it does not have a splined shaft as it is driven by pitman arm 350. Rather, pitman arm 420 includes a cylindrical bearing portion 422 having a through opening at 424. Pitman arm 420 also includes an arm portion 428 having apertures 430 and 432. As shown best in FIG. 18, bracket 112 includes a portion 440 (FIGS. 18-19), which is coupled to tubes 102 having upper plate portion 442 having an aperture 444 aligned with aperture 446 of lower plate portion 400. Thus, as shown in FIGS. 19, 20, driven pitman arm 420 can be coupled between bracket portions 112, 304 by way of a fastener 450 extending through aperture 326 of tab portion 322 extending downwardly through aperture 424 of driven pitman arm 420 and through apertures 444, 446 of bracket portion 112. A complementary fastener 452 may be coupled to fastener 450 to retain driven pitman arm 420 in place.

With reference still to FIGS. 18 and 19, drive and driven pitman arms 350, 420 are coupled together by way of a drag link 460, where drag link 460 includes a bar portion 462 having couplings 464 at each end. Each coupling 464 includes an aperture 466 which align with apertures 362 and 430 to receive fasteners 470 therethrough. Fasteners 470 couple with fasteners 472 to retain drag link 460 in a coupling relationship with the drive pitman arm 350 and driven pitman arm 420, as shown best in FIG. 19.

Finally, with respect to FIG. 18, steering arms 222 each include a ball joint 500 having a threaded end 502, where each of the threaded ends 502 is received in one of the apertures 364 or 432. Thus, as should be appreciated, input from the splined shaft 314 to drive pitman arm 350 causes a like rotation of the driven pitman arm 420 by way of drag link 460, which in turn causes transverse movement of the steering arms 222 in a push-pull relationship with the pitman arms 350, 420 depending upon the direction of rotation.

With reference now to FIGS. 11 and 16, further features of the steering system will be described. As shown best in FIG. 11, the steering shaft 300 extends along a non-vertical axis 510 with respect to vertical axis 512 to provide an angular offset β_1 , where β_1 =approximately 8.5°, but could be in a range from 5°-12°, and more particularly from 7°-10°. This positions the center of the left inner ball joint 500 (FIG. 16) at a left vertical position 516. As shown in

FIG. 11, vertical position 516 is offset from the vertical center line 520 and the vertical center 522 of the right inner ball joint 500 is laterally offset from vertical center line 520 as well. As shown, left inner ball joint 500 is offset from the vertical center line 520 by a distance of 526, whereas the right inner ball joint 500 is offset from the vertical center line 520 by a distance of 528, where distances 526 and 528 are equal, and are equal to 3.25", but could be in a range from 2"-5". Thus, the laterally offset steering motor in combination with the drag link 460 eliminates the bump steer that is introduced into the suspension.

Furthermore, with reference to FIGS. 16 and 19, each of the arm portions 360 and 428 of respected pitman arms 350 and 420 include portions skewed at 360a and 428a relative to a horizontal axis 530 such that portions 462a and 462b of drag link 460 form angles β_2 relative to horizontal axis 530, where $\beta_2=15^\circ$, but could be in a range from 10° - 20° .

With reference now to FIGS. 22 and 23, the front suspension of the 48" and 55" vehicles will be distinguished. With reference first to FIG. 22, ATV 2 is in the form of the 55" vehicle where a distance from outer faces of wheel hubs 240 is shown by a lateral distance X_1 , where $X_1=49"$. In this way, the distance X_1 plus the width dimensions of the tires would approximate the 55" width vehicle. It should also be understood that the suspension as described with respect to FIGS. 10-16, depict the 55" vehicle such that the upper and lower A-arms 200, 202; 204, 206 relate to the 55" vehicle. The half shafts 256 are also of a different length to accommodate the wider vehicle. As shown in FIG. 22, the wheel hubs 240 are positioned further forward relative to the frame 20 such that the half shafts 256 sweep forward relative to a horizontal center line 540 through the couplings 260. As shown, the half shafts 256 are swept forward by an angle β_3 , where $\beta_3=2.8^\circ$, but could be in a range from 0° - 5° .

With reference now to FIG. 23, the front suspension for the 48" vehicle is shown at 56', where the front suspension 56' includes left A-arms 200' and 202' and right A-arms 204' and 206'. The half shafts 256' are also of a different length to accommodate the lateral difference between the couplings 260 and the hubs 240. As shown, the lateral distance between front faces of the hubs 240 is X_2 , such that the distance X_2 plus the width of the wheels equates to the 48" vehicle. As shown, $X_2=42"$. As also shown in FIG. 23, the half shafts 256' are swept rearwardly relative to the lateral center line 540 such that an angle β_4 is defined which reduces the wheel base of the vehicle relative to the 55" vehicle described in FIG. 22. As shown, the steering system 36 of the 48" and 55" vehicle are identical with the exception that the steering arms 22' of the 48" vehicle are somewhat shorter to accommodate the width difference of the vehicles. As also shown, $\beta_4=7.4^\circ$, but could be in a range from but could be in a range from 5° - 10° .

With reference now to FIGS. 24 and 25, the front suspension of the 48" and 55" vehicles will be described. With reference first to FIG. 24, the 55 inch vehicle is shown where the front left wheel is turned to the maximum left position, to a rotation of $\beta_5=35^\circ$. Due to the fact that the half shafts 256 sweep forward on the 55" vehicle (FIG. 22) the first contact between the CV joint 250 (FIG. 13) and the half shaft 256, would be at the rearward side of the CV joint 250. Thus the upstanding post 284 on the 55" vehicle contacts the rear contact surface 270 to form the extreme position for the 55" vehicle.

With reference now to FIG. 25, the 48 inch vehicle is shown where the front left wheel is turned to the maximum right position, to a rotation of $\beta_6=35^\circ$. Due to the fact that the half shafts 256' sweep rearward on the 48" vehicle (FIG. 23)

the first contact between the CV joint 250 (FIG. 13) and the half shaft 256', would be at the forward side of the CV joint 250. Thus the upstanding post 284 on the 48" vehicle contacts the front contact surface 268 to form the extreme position for the 48" vehicle.

It should also be understood that the knuckles 224, hubs 240 and the couplings 276 are identical for the 48" and 55" vehicles and that the stops 268, 270 are defined for providing the extreme position for both vehicles.

With reference now to FIGS. 26-31, the rear suspension 58 of the ATV 2 will be described. With reference first to FIGS. 24 and 25, the suspension of the 55" vehicle will be described. As shown, the suspension 58 is comprised of a left lower A-arm 550, a left upper A-arm 552, a lower right A-arm 554 and a right upper A-arm 556. The left A-arms 550, 552 are coupled to a knuckle 560 which retains a left rear wheel hub 562. Wheel hub 562 is driven by a rear left half shaft 564. The right A-arms 554 and 556 are coupled to a right rear knuckle 570 and a right rear wheel hub 572 where wheel hub 572 is driven by a right rear half shaft 574. A torsion bar 580 couples the lower A-arms 550, 554 by way of vertical posts 582. As shown, half shafts 564, 574 are driven by couplings 586 of a final drive 590.

With reference now to FIGS. 28 and 29, the rear suspension 58' for the 48" vehicle will be described. As with the front suspension, the majority of the components of the vehicle remain the same, but the left A-arms 550', 552' and half shaft 564' are replaced to compensate for the vehicle width. Similarly, the right-hand side suspension is also changed with lower A-arm 554', upper A-arm 556' and half shaft 574' having the reduced dimensions like the left-hand side. The torsion bar 580' is also different to accommodate the width of the vehicle.

With reference now to FIG. 28, the coupling between the A-arms and the knuckle will be described. As shown in FIG. 28, the right rear suspension is shown where A-arms 554 and 556 are coupled to knuckle 570. This is also the same knuckle and wheel hub used on the 48" vehicle and mirror images of the left-hand side wheel hub. The knuckle 570 includes a single elongate coupling 590 at the upper end thereof for coupling to upper A-arm 556 and includes two lower couplings at 592 for attachment of A-arm couplings 594. Wheel hub 572 includes a constant velocity joint 600 extending through the knuckle 570 which includes a drive profile 602 for matching with a constant velocity joint on half shaft 574. As shown in FIG. 28, the coupling points 592 are raised relative to the constant velocity coupling 600 such that the top of the coupling points 594 are higher than the inner geometry 602.

With reference now to FIG. 31, the rear final drive 590 is shown mounted in the rear frame portion 72 positioned above frame tubes 140 and below frame tubes 142. Rear suspension mounts 150, 166 and 170 are shown in greater detail. Final drive 590 is shown positioned adjacent to forging 82 which may be removed to access the final drive 590.

With respect now to FIGS. 32-35, the exhaust system 52 will be described in greater detail. As shown, exhaust system 52 includes an exhaust pipe 610 coupled to engine 612 at a front end thereof, and extending rearwardly past continuously variable transmission (CVT) 614 to an elbow 616, and then to a front end of muffler 620. As shown, muffler is centered in the vehicle frame 20 above tube 146, intermediate tubes 80c and below tubes 90. As shown best in FIG. 32, muffler 620 includes a muffler mounting post 630, having post portions 632 extending into apertures 174 (FIG. 35) of muffler mount 172. The muffler 620 is centrally

mounted because with the extended travel of suspension the muffler **620** cannot be packaged in the wheel well at full rebound/jounce.

It should be understood that the vehicle **2** includes front and rear drive shafts coupled between U-joint **634** of front drive **190** (FIG. **15**) and output U-joint **635** (FIG. **32**); and between U-joint **636** of rear drive **590** (FIG. **31**) and output U-joint **638** (FIG. **32**). The drive shafts are similar to those described in U.S. Pat. No. 9,873,316, the subject matter of which is incorporated herein by reference.

With reference now to FIGS. **36-38**, an alternate steering system is shown as a rack and pinion system **640** which is an alternative to the steering system shown in FIG. **18**. Rack and pinion system **640** utilizes the power steering motor **302** as disclosed in the embodiment of FIG. **18**, but couples to it a rack and pinion subassembly **642**. As shown, subassembly **642** includes a housing **644** having bearing covers **646** and **648** housing bearings **650**, **652**. Bearings **650** and **652** support a stub shaft **656** which includes a splined internal shaft which couples to splined drive shaft **314** of power steering motor **310**. A pinion **660** is coupled to stub shaft **656**, and as best shown in FIG. **38**, pinion **660** has an arcuate shape at **662** having gear teeth which mesh with teeth **664** of a rack **666**. The rack includes couplers **668** which couple to steering arms **222** (FIG. **18**) as opposed to the utilization of pitman arms **350**, **420**. Thus, as shown in FIG. **38**, when pinion **660** rotates in a clockwise direction, rack **666** moves linearly in the direction of arrow **670** and when pinion **660** rotates in a counterclockwise direction, rack **666** moves linearly in the direction of arrow **672**.

With reference now to FIGS. **39** and **40**, a steering riser block assembly **680** and an adjustable instrument cluster **682** will be described herein. As shown, riser block assembly **680** is coupled to an alternate steering arm **684** having a T-shaped post at **686**. Riser block assembly **680** includes the main body portion **690**, which in the disclosed embodiment is an aluminum block extruded in the direction of apertures **692** in post **694**. Block **690** includes an upper surface **696** and a lower surface **698** which receives upper clamps **700** and lower clamps **702** respectively. Lower clamps **702** include an arcuate opening at **706** for placement against T-shaped post **686** whereby fasteners **708**, **710** may be utilized to attach clamps **702** to bottom surface **698** trapping T-shaped post **686** therebetween. Lower end of main body portion **690** includes an arcuate opening at **712** to receive the T-shaped post **686**. It should be appreciated that a lower end of aperture **692** is threaded, to receive threads on fasteners **708**.

As shown best in FIG. **40**, instrument cluster **682** includes a bracket **716** including a lower plate portion **718** having apertures **720**. Apertures **720** align with apertures **722** of upper clamps **700**. Thus, fasteners **726** may be received through apertures **720** of bracket **716**, through apertures **722** of clamps **700** into the upper end of apertures **692**. It should be appreciated that the upper ends of apertures **692** are also threaded to receive threads on fasteners **726**. Thus, when bracket **716** is coupled to riser block **690**, handlebar **740** is coupled therebetween and upper clamps **700** and riser block **690** grips a knurled surface **742** to retain the position of the handlebars **740**. Upper end on main housing **690** includes an arcuate opening **744** and clamp **700** includes an arcuate opening **745** to grip knurled surface **742**. Bracket **716** includes a plurality of bracket arms **746**, **748**, **750** which allows coupling of upper housing **760** of pod **682** by way of fasteners **762** to bracket **716**.

Thus, given the geometry of the riser block **690**, the riser block **690** may rotate towards and away from the rider on

T-shaped post **686** which moves handlebars **740** towards or away from the rider to accommodate the ergonomic feel for different sized riders. This also adjusts the pod **682** to accommodate different sized riders. Furthermore, the riser block **690** allows the handlebars **740** to be rotated downwardly in a stored or shipped position as shown in the phantom lines of FIG. **39**.

While this invention has been described as having an exemplary design, the present invention may be further modified within the spirit and scope of this disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the invention using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this invention pertains.

What is claimed is:

1. An all-terrain vehicle (ATV), comprising:
 - a frame;
 - ground engaging members supporting the frame, including front and rear ground engaging members;
 - front right and left suspensions, at least one of the right and left suspensions including an upper arm and a lower arm;
 - front wheel steering mechanisms including right and left knuckles pivotally coupled to the respective front right and left suspensions;
 - a power steering unit coupled to the frame, the power steering unit including an output shaft which is coupled to the right and left knuckles for steering;
 - a first stop member positioned on each of the front right and left suspension, the first stop member defined on the lower arm; and
 - a second stop member positioned on each of the right and left knuckles and configured to receive a portion of the first stop member therein,
 wherein the first and second stop members cooperate to provide extreme rotational positions for the power steering unit, wherein the extreme rotational positions are defined by the knuckle.
2. The all-terrain vehicle of claim 1, wherein the first stop member includes an upstanding post.
3. The all-terrain vehicle of claim 2, wherein the upstanding post has a shark fin geometry.
4. The all-terrain vehicle of claim 1, wherein the first stop member is configured to contact the second stop member at a 45-degree angle.
5. The all-terrain vehicle of claim 1, wherein the second stop member defines an arcuate slot.
6. The all-terrain vehicle of claim 1, wherein the knuckle defines a pin and the lower arm defines an opening, wherein the pin is configured to be received within the opening to couple the lower arm to the knuckle.
7. The all-terrain vehicle of claim 6, wherein the lower arm is rotatable about the pin and the lower arm is configured to rotate between the extreme rotational positions.
8. A suspension structure of an all-terrain vehicle comprising:
 - a shock absorber;
 - an upper arm and a lower arm operatively coupled to the shock absorber;
 - a knuckle coupled between the upper arm and the lower arm;
 - a stopper defined by the knuckle and configured to regulate a displacement angle between the lower arm and the knuckle, the stopper defining an arcuate slot; and

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a projection defined by the lower arm and configured to abut surfaces of the stopper during rotation.

9. The suspension structure of claim 8, wherein the projection defines a top and a base opposite the top, wherein a width of the projection at the top is less than a width of the projection at the base.

10. The suspension structure of claim 9, wherein the top of the projection is configured to abut the surfaces of the stopper during rotation.

11. The suspension structure of claim 8, wherein the stopper includes a first end and a second end opposite the first end, and the surfaces of the stopper include a first extreme contact surface at the first end and a second extreme contact surface at the second end.

12. An all-terrain vehicle comprising:

a frame;

ground engaging members supporting the frame, including front and rear ground engaging members;

a front suspension including an upper arm and a lower arm;

a front wheel steering mechanism including a knuckle pivotably coupled to the front suspension;

a power steering unit coupled to the frame and configured for steering, the power steering unit including an output shaft which is coupled to the knuckle;

a projection member defined by the lower arm; and

a stopper defined by the knuckle and configured to regulate a displacement angle between the lower arm and the knuckle,

wherein the stopper defines an arcuate slot, and

wherein the projection is configured to abut surfaces of the stopper during rotation of the power steering unit.

13. The all-terrain vehicle of claim 12, wherein the projection defines a top and a base opposite the top, wherein a width of the projection at the top is less than a width of the projection at the base.

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14. The all-terrain vehicle of claim 12, wherein the knuckle is operatively coupled to a wheel hub of one of the front ground engaging members.

15. The all-terrain vehicle of claim 12, further including a steering arm coupled to the knuckle and positioned between the upper arm and the lower arm.

16. The all-terrain vehicle of claim 15, wherein the knuckle defines a top end, a bottom end opposite the top end, and a length therebetween, and wherein the knuckle includes an extension defined along the length and the steering arm is coupled to the extension.

17. The all-terrain vehicle of claim 15, wherein the upper arm has a first end, a second end, and a length therebetween, and the upper arm defines an upwardly directed bend along a portion of the length configured to provide clearance to the steering arm.

18. The all-terrain vehicle of claim 12, wherein the lower arm has a first end, a second end, and a length therebetween, wherein the lower arm is curved along the length.

19. A suspension structure of an all-terrain vehicle comprising:

a shock absorber;

an upper arm and a lower arm operatively coupled to the shock absorber;

a knuckle coupled between the upper arm and the lower arm;

a stopper defined by the knuckle and configured to regulate a displacement angle between the lower arm and the knuckle, wherein the stopper includes a first end and a second end opposite the first end, and the surfaces of the stopper include a first extreme contact surface at the first end and a second extreme contact surface at the second end; and

a projection defined by the lower arm and configured to abut surfaces of the stopper during rotation.

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